
Generalized Flow-Matching-Guided Framework for Tabular Data Generation

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Generating tabular data that satisfies user-specified constraints—value ranges, cat-
2 egorical requirements, and logical combinations—without retraining remains a
3 significant challenge. Existing DDPM-based approaches can achieve training-
4 free constrained generation but suffer from degraded fidelity on heterogeneous
5 features; flow matching methods attain state-of-the-art unconditional quality but
6 lack inference-time conditioning. We present a unified framework that ex-
7 tends manifold-guided generation from DDPM to the Bregman geometry of
8 exponential-family variational flow matching (EF-VFM), introducing a Guided
9 Euler Sampler that applies x_t -space gradient corrections at inference time—
10 requiring zero additional training while benefiting from the faster sampling of flow
11 matching. We provide theoretical justification by showing that mean-field and
12 joint Bregman losses yield identical loss landscapes under product exponential
13 families, and that categorical features commit at rate $O(e^{-\alpha/4})$ —exponentially
14 faster than continuous features at $O(\alpha^{-2})$ —during generation. Empirically, the
15 framework achieves constraint violation rates substantially lower than those of ex-
16 isting baselines across multiple constraint types, while maintaining state-of-the-art
17 unconditional fidelity and competitive imputation performance.

18 1 Introduction

19 Synthetic tabular data is increasingly demanded in healthcare [1, 2], finance [3, 4], and privacy-
20 preserving data sharing [5, 6]. As these demands grow, constrained tabular generation—producing
21 samples that satisfy user-specified conditions—has become a pressing research imperative. Down-
22 stream applications span missing-value imputation under partial observations [7–11] and what-if
23 scenario simulation for decision-making [1, 12]. Deep generative models for tabular data have
24 progressed from GAN-based approaches [13], through diffusion models [14–19], to flow matching
25 methods [20, 21], with each generation improving unconditional fidelity. Yet extending these models
26 to constrained settings remains challenging: most conditional generation methods embed constraint
27 information during training, preventing generalization to new constraint types at inference time;
28 switching constraints requires retraining or fails to handle diverse constraint targets efficiently [1].

29 A notable exception is Harpoon [8], which achieves training-free constrained generation within
30 DDPM by proving that the denoiser acts as an orthogonal projection onto the data manifold. Since
31 constraints define subregions of the manifold, applying gradient corrections along the manifold
32 surface steers samples toward the feasible region, while the manifold’s low-dimensional structure
33 ensures that corrected dimensions pull unconstrained ones into consistency. However, Harpoon’s
34 unified Gaussian noise paradigm limits fidelity on heterogeneous tabular features, and its DDPM
35 backbone incurs higher sampling cost than flow matching. Migrating this principle to flow match-
36 ing is non-trivial: flow matching uses OT interpolation rather than Gaussian diffusion, yielding a

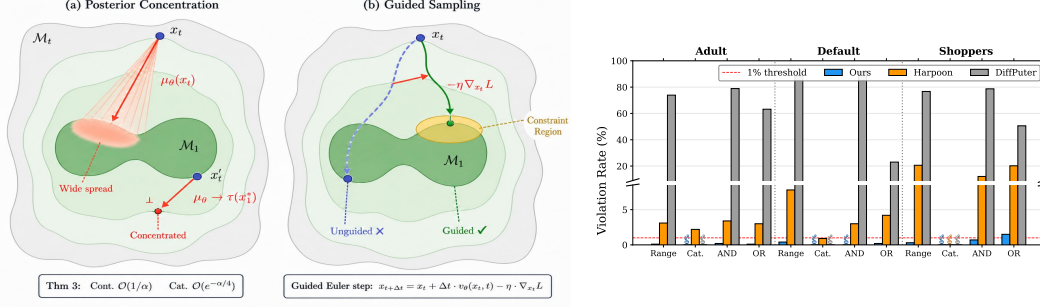


Figure 1: The predictor $\mu_\theta(x_t)$ concentrates onto the data manifold as $t \rightarrow 1$, with continuous error $O(1/\alpha)$ and categorical error $O(e^{-\alpha/4})$ (left). The gradient correction $-\eta \nabla_{x_t} \mathcal{L}_{\text{inf}}$ steers trajectories toward the feasible region. Right: our method achieves near-zero constraint violation across 3 datasets and 4 constraint types while baselines exhibit 3–91% failure rates.

different predictor geometry—the Bregman sufficient-statistic predictions of EF-VFM [20] replace DDPM’s denoiser, requiring new theoretical grounding for manifold-guided generation.

We address this challenge by establishing that EF-VFM’s Bregman predictor concentrates onto the data manifold as $t \rightarrow 1$, providing the geometric foundation needed for gradient-based guidance within the Bregman framework. We further confirm that the mean-field factorization used in EF-VFM training incurs no loss relative to joint training (Free Lunch Theorem), and characterize a two-phase generation dynamic where categorical features commit exponentially faster than continuous ones (TC Decay Theorem). Building on these results, we design the Guided Euler Sampler, which injects x_t -space gradient corrections at each Euler step to enforce constraints at inference time—requiring zero retraining. Our contributions are summarized as follows:

Theory. We prove the Free Lunch Theorem: mean-field and joint Bregman losses are identical under product exponential families (Theorem 1). We establish the TC Decay Theorem: categorical conditional entropy decays at $O(e^{-\alpha/4})$ while continuous mutual information decays at $O(\alpha^{-2})$, revealing a two-phase generation dynamic that informs sampler design (Theorem 2).

Algorithm. Based on these theoretical results, we design the Guided Euler Sampler, a manifold-aware inference-time conditioning method for flow matching that unifies inequality constraints, categorical controls, logical combinations, and missing-value imputation through gradient-guided Euler integration—requiring zero retraining.

Empirical validation. We empirically validate the geometric alignment between guidance gradients and the data manifold, and demonstrate that the Guided Euler Sampler achieves constraint violation rates over an order of magnitude lower than existing baselines across 7 datasets while maintaining competitive imputation performance.

2 Related Work

We survey related work along two axes: deep generative models for tabular data with their conditional extensions, and manifold-guided generation.

2.1 Deep Tabular Models and Conditional Extensions

Deep generative models for tabular data have evolved from GAN-based methods [13], through diffusion models [14–17, 19], to flow matching approaches [20–24]. For conditional generation, most methods require training-time conditioning [12, 19, 25]. For imputation, dedicated architectures integrate diffusion with EM, masked score matching, deterministic conditional diffusion, or masked autoencoders [7, 11, 18, 26]. Inference-time methods—classifier guidance, RePaint [27, 28]—avoid retraining entirely but remain confined to DDPM backbones. Training-time approaches lack generalization to unseen constraint types, while existing inference-time methods have not addressed the heterogeneous feature structure of tabular data under flow matching. Our work extends inference-time

71 constrained generation to the flow matching framework with native support for mixed continuous-
72 categorical features.

73 2.2 Manifold Guidance

74 The manifold hypothesis—that high-dimensional data concentrates near low-dimensional
75 submanifolds—underpins inference-time guidance for diffusion models. In the image domain,
76 DPS [31] applies likelihood gradients at each reverse step, and LGD [34] provides plug-and-play
77 loss-guided sampling via Monte Carlo corrections. Harpoon [8] extends manifold guidance to tabu-
78 lar DDPM, proving that the denoiser acts as an orthogonal projector onto the data manifold. How-
79 ever, all these methods operate within the DDPM Gaussian-noise framework; Harpoon further relies
80 on unified Gaussian noise for all features, limiting fidelity on heterogeneous columns. This paper
81 addresses this challenge by adopting the Bregman geometry of EF-VFM, whose product exponen-
82 tial family structure [20, 35, 36] provides a natural fit for heterogeneous tabular features, enabling
83 type-adapted gradient corrections that overcome the geometric mismatch between DDPM and flow
84 matching.

85 3 Preliminaries

86 This section establishes the mathematical foundations for the proposed framework. We first re-
87 view the transport framework and flow matching (§3.1), then introduce exponential families and
88 the variational flow matching formulation that connects them (§3.2). Finally, we summarize the
89 manifold-guided inference paradigm from Harpoon (§3.3), motivating the extensions developed in
90 Section 4.

91 3.1 Flow Matching and OT Interpolation

92 Modern generative modeling interprets sampling from a target distribution p_1 as transporting a base
93 distribution p_0 along a continuous time path [37]. A time-dependent mapping $\phi_t : [0, 1] \times \mathbb{R}^D \rightarrow \mathbb{R}^D$
94 evolves samples via the ODE

$$\frac{d}{dt}\phi_t(x) = u_t(\phi_t(x)), \quad \phi_0(x) = x, \quad (1)$$

95 where u_t is a time-dependent velocity field and ϕ_1 pushes p_0 onto p_1 . Flow matching [37] cir-
96 cumvents the cost of solving ODEs during training by regressing the velocity field directly via the
97 conditional flow matching (CFM) objective:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, x_1, x} [\|u_t(x | x_1) - v_\theta(x, t)\|^2], \quad (2)$$

98 where $u_t(x | x_1)$ is the conditional velocity pointing toward endpoint x_1 . Under OT linear in-
99 terpolation, $x_t = (1 - t)\varepsilon + tx_1$ with $\varepsilon \sim \mathcal{N}(0, I)$ and $x_1 \sim p_1$, this conditional velocity is
100 $u_t(x | x_1) = (x_1 - x)/(1 - t)$. Samples are generated by integrating $dx_t/dt = v_\theta(x_t, t)$ from
101 $x_0 \sim \mathcal{N}(0, I)$.

102 In practice, a softened interpolation $x_t = (1 - \sigma t)\varepsilon + \sigma t x_1$ with $\sigma < 1$ (e.g., $\sigma = 0.99$) is used
103 to avoid the singularity at $t = 1$ [20]. When the model predicts the mean parameter $\mu_\theta(x_t) \approx$
104 $\mathbb{E}[\tau(x_1) | x_t]$ rather than the velocity directly, the velocity is recovered as

$$v_\theta(x_t, t) = \frac{\mu_\theta(x_t) - \sigma x_t}{1 - \sigma t}. \quad (3)$$

105 **Variational Flow Matching.** The unknown posterior $p_t(x_1 | x)$ can be replaced by a variational
106 distribution $q_\theta^t(x_1 | x)$, yielding the VFM objective [38]:

$$\mathcal{L}_{\text{VFM}}(\theta) = -\mathbb{E}_{t, x_1, x} [\log q_\theta^t(x_1 | x)]. \quad (4)$$

107 When the conditional velocity is linear in x_1 , the marginal velocity depends only on $\mathbb{E}[x_1 | x_t]$. A
108 mean-field factorization $q_\theta^t(x_1 | x) = \prod_d q_\theta^d(x_1^d | x)$ then suffices, reducing the problem to learning
109 D univariate distributions [20].

3.2 Exponential-Family Variational Flow Matching

The mean-field VFM formulation requires choosing a distribution family for each $q_\theta^t(x_1^d | x)$. Exponential families provide a unified choice: a distribution belongs to the exponential family [36] if it can be written as

$$p(x | \eta) = h(x) \exp(\tau(x) \cdot \eta - A(\eta)), \quad (5)$$

where η are natural parameters, $\tau(x)$ are sufficient statistics, and $A(\eta)$ is the log-partition function. Mean parameters $\mu = \mathbb{E}[\tau(x)] = \nabla A(\eta)$ provide an equivalent parameterization via the conjugate dual A^* , with $\eta = \nabla A^*(\mu)$.

When each $q_\theta^t(x_1^d | x)$ belongs to an exponential family, the VFM loss is equivalent to minimizing a Bregman divergence [35] induced by the conjugate dual:

$$D_{A^*}(u, v) = A^*(u) - A^*(v) - \langle u - v, \nabla A^*(v) \rangle. \quad (6)$$

For Gaussians, D_{A^*} reduces to $\frac{1}{2}\|u - v\|^2$; for categoricals, it becomes $\text{KL}(u\|v)$. Under a product exponential family $p(x | \eta) = \prod_d p_d(x^d | \eta^d)$, the conjugate decomposes additively as $A^* = \sum_d A_d^*$, and the training objective becomes the mean-field Bregman loss:

$$\mathcal{L}(\theta) = \sum_{d \in \mathcal{C}} D_{A_d^*}(\tau_d(x_1^d), \mu_t^{\theta, d}(x_t)) + \sum_{d \in \mathcal{D}} D_{A_d^*}(e_{x_1^d}, \text{softmax}(\text{logits}_t^{\theta, d}(x_t))), \quad (7)$$

where $\mathcal{C}$ and $\mathcal{D}$ index continuous and discrete dimensions, respectively. The population-optimal predictor is the Bregman centroid [35]: $\mu^{*, d}(x_t) = \mathbb{E}[\tau_d(x_1^d) | x_t]$. In our implementation, a neural network $f_\theta(x_t, t)$ directly outputs these per-dimension sufficient statistics—mean predictions μ_θ^d for continuous features and classification logits for categorical features—and is trained by minimizing $\mathcal{L}(\theta)$ in Eq. (7).

3.3 Manifold-Guided Inference

Harpoon [8] establishes that, under the Gaussian noising framework of DDPM, the denoiser’s dirty estimate $Q_t(x_t) := \frac{1}{\sqrt{\alpha_t}}(x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t, t))$ converges to the orthogonal projection $\pi(x_t)$ onto the data manifold $\mathcal{M}_0$ as $\alpha_t \rightarrow 1$. Formally, for any differentiable inference-time loss $\mathcal{L}_{\text{inf}}$, the gradient $\nabla_{x_t} \mathcal{L}_{\text{inf}}(Q_t(x_t))$ lies in the tangent space $T_{\hat{x}_0} \mathcal{M}_0$ [8]. The guided reverse process takes the form

$$x_{t-1} = \text{Denoise}(x_t) - \eta \nabla_{x_t} \mathcal{L}_{\text{inf}}(Q_t(x_t)), \quad (8)$$

where η controls guidance strength. This framework enables inference-time conditioning without retraining, but operates within DDPM and does not extend to the Bregman geometry of EF-VFM. Migrating this paradigm to flow matching while preserving theoretical guarantees is the goal of our method (Section 4).

4 Method

Our model is trained with the EF-VFM objective of Eq. (7); Theorems 1 and 3 (Section 5) show that this mean-field loss is lossless and that the trained predictor $\mu_\theta(x_t)$ converges to the ground-truth sufficient statistics as $t \rightarrow 1$. We now describe how to exploit this predictor for inference-time constraint enforcement: Section 4.1 constructs the Guided Euler Sampler that migrates Harpoon’s manifold guidance (§3.3) to the Bregman geometry of flow matching; Section 4.2 defines the family of constraint losses.

4.1 Guided Euler Sampler

Section 3.3 showed that Harpoon’s denoiser estimate Q_t converges to the manifold projection in DDPM (Eq. (8)). In EF-VFM, the Bregman predictor $\mu_\theta(x_t)$ plays an analogous role: Posterior Concentration (Theorem 3) guarantees $\mu_\theta(x_t) \rightarrow \tau(x_1^*)$ as $t \rightarrow 1$, enabling accurate gradient-based guidance.

Without guidance, generation integrates the learned velocity field (Eq. (3)) via standard Euler steps. To enforce constraints without retraining, we augment each Euler step with a gradient correction

Table 1: Inference-time loss functions for constrained generation and imputation. Fairness is a direct application of the AND loss to sensitive attributes (Appendix H.13).

Type	$\mathcal{L}_{\text{inf}}$	Description	Example
Range	$\text{ReLU}(l - \mu_j) + \text{ReLU}(\mu_j - u)$	Bound continuous $\in [l, u]$	<code>age</code> $\in [25, 50]$
Category	$-\log \text{softmax}(\text{logits}_j)_k$	Fix discrete to target k	<code>occupation=Eng.</code>
AND	$\sum_i \mathcal{L}_i$	All conditions satisfied	<code>age</code> $\in [25, 50] \wedge \text{edu} = \text{PhD}$
OR	$\min_i \mathcal{L}_i$	At least one satisfied	<code>age</code> $< 30 \vee \text{income} > 50\text{K}$
Imputation	RePaint + optional $\nabla \mathcal{L}$	Replace observed dims	Fill missing values
Fairness	$\mathcal{L}_{\text{AND}}$ on sensitive attr.	Demographic subgroup	<code>sex=F</code> $\wedge$ <code>inc</code> $> 50\text{K}$

151 toward constraint-satisfying regions:

$$x_{t+\Delta t} = x_t + \Delta t \cdot \underbrace{\frac{\mu_\theta(x_t) - \sigma x_t}{1 - \sigma t}}_{\text{base velocity}} - \eta \nabla_{x_t} \mathcal{L}_{\text{inf}}(\mu_\theta(x_t)), \quad (9)$$

152 where η is the guidance strength and $\mathcal{L}_{\text{inf}}$ is a differentiable loss (§4.2). The gradient propagates
153 through μ_θ via the chain rule.

154 Crucially, the guidance term in Eq. (9) is *not* scaled by Δt . This makes the sampler an interleaved
155 projection scheme rather than a modified ODE discretization, ensuring that guidance strength re-
156 mains robust regardless of the integration step size (Appendix C.5).

157 This formulation justifies several design choices validated in our ablation (§6.4). By operating in
158 x_t -space, the gradient propagates through the model’s Jacobian J_{μ_θ} , automatically mediating hetero-
159 geneous feature scales without requiring per-type guidance strengths. The Euler solver is necessary
160 because it applies the correction immediately after every velocity evaluation, whereas higher-order
161 methods evaluate unguided velocities multiple times per correction. Finally, a constant η suffices be-
162 cause the inherent two-phase generation dynamic (Theorem 2) naturally decays categorical guidance
163 gradients once those features commit.

164 4.2 Inference-Time Loss Functions

165 The modular design of the Guided Euler Sampler reduces the problem of supporting diverse con-
166 straint types to designing appropriate differentiable loss functions. Each constraint is expressed as
167 a loss $\mathcal{L}_{\text{inf}}$ that measures deviation from the feasible region, and the sampler enforces it through
168 gradient descent in x_t -space.

169 The Guided Euler Sampler supports a family of differentiable losses $\mathcal{L}_{\text{inf}}$, summarized in Table 1.
170 Range constraints bound a continuous feature within a user-specified interval (e.g., generating in-
171 dividuals aged 25–50). Category constraints fix a discrete feature to a target value (e.g., requiring
172 `occupation = Engineer`). AND constraints enforce the simultaneous satisfaction of multiple
173 conditions, creating a tighter feasible region (e.g., `age` $\in [25, 50]$ *and* `education = PhD`). OR
174 constraints require at least one condition to hold, yielding a broader feasible set (e.g., `age` < 30 *or*
175 `income` $> \$50\text{K}$). Imputation replaces observed dimensions via RePaint [28] with optional gradient
176 refinement.

177 The complete procedure is summarized in Algorithm 1.

178 5 Theoretical Analysis

179 We establish three results for the EF-VFM framework of §3. Detailed proofs appear in Appendix B.

180 The EF-VFM loss (Eq. 7) uses the product exponential family parameterization $A^* = \sum_d A_d^*$, under
181 which the Bregman divergence decomposes additively. By Fubini’s theorem, the expectation and the
182 per-dimension sum commute:

$$\mathcal{L}_{\text{joint}} = \mathbb{E}[D_{A^*}(\tau(x_1), \mu_\theta(x_t))] = \sum_d \mathbb{E}[D_{A_d^*}(\tau_d(x_1^d), \mu_\theta^d(x_t))] = \mathcal{L}_{\text{MF}}. \quad (10)$$

Algorithm 1 Guided Euler Sampling

Require: Trained EF-VFM model f_θ , guidance strength η , steps N , loss $\mathcal{L}_{\text{inf}}$, optional mask $\mathbf{m}$ and observed x_{obs}

Ensure: Generated sample x_1 satisfying constraints

```

1: Sample  $x_0 \sim \mathcal{N}(0, I)$ ; set  $\Delta t = 1/N$ 
2: for  $n = 0, 1, \dots, N - 1$  do
3:    $t \leftarrow n \cdot \Delta t$ 
4:    $\mu_\theta \leftarrow f_\theta(x_t, t)$  {Predict Bregman sufficient statistics}
5:    $v_t \leftarrow (\mu_\theta - \sigma x_t)/(1 - \sigma t)$  {Base velocity (Eq. (3))}
6:    $g_t \leftarrow \nabla_{x_t} \mathcal{L}_{\text{inf}}(\mu_\theta)$  {Guidance gradient}
7:    $x_{t+\Delta t} \leftarrow x_t + \Delta t \cdot v_t - \eta \cdot g_t$  {Guided Euler step}
8:   if imputation then
9:      $x_{t+\Delta t}[\mathbf{m}] \leftarrow (1 - (t+\Delta t))\varepsilon[\mathbf{m}] + (t+\Delta t)x_{\text{obs}}[\mathbf{m}]$ 
10:  end if
11: end for
12: return  $x_1 \leftarrow x_{N \cdot \Delta t}$ 
  
```

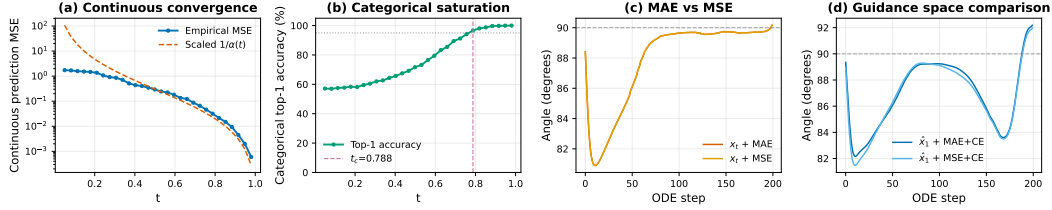


Figure 2: Posterior concentration on Adult: continuous MSE follows $O(1/\alpha)$ and categorical accuracy reaches 99% at $t_c=0.867$, close to the theoretical 0.821. Guidance gradients remain near-orthogonal ($\approx 89^\circ$) to the base velocity throughout sampling. Shaded: ± 1 std, 5 seeds.

183 Since this identity holds for all θ , the two losses share the same minimizer μ^* . Substituting μ^*
 184 into the velocity formula (Eq. 3) gives $v_{\text{MF}}^d = v_{\text{joint}}^d$ for every dimension d —that is, training per-
 185 dimension losses independently recovers the exact same velocity field as joint training. By Picard–
 186 Lindelöf uniqueness, the ODE trajectories and hence the generated distributions coincide.

187 **Theorem 1** (Free Lunch). *Under product exponential family output ($A^* = \sum_d A_d^*$), the mean-field*
 188 *and joint velocity fields are identical at the population optimum for all d, t , and both produce the*
 189 *same transport map $T_{\#}\mathcal{N}(0, I) = p_1$.*

190 A direct consequence is that the optimality gap $\text{TC} = \mathcal{L}_{\text{MF}}^* - \mathcal{L}_{\text{joint}}^* = 0$ at the population level: per-
 191 dimension training incurs no quality penalty. This extends the score matching equivalence of [39]
 192 from continuous Gaussian models to mixed-type Bregman losses. Proofs, a Grönwall finite-sample
 193 bound, and necessity of additive A^* are in Appendix B.1 and C.1.

194 Having shown that training is lossless, we next analyze the sampling phase: how does the ODE
 195 trajectory resolve individual features? Under OT interpolation $x_t = tx_1 + (1-t)\varepsilon$, define the signal-
 196 to-noise ratio $\alpha(t) = t^2/(1-t)^2$. As α grows, the posterior $p(x_1 | x_t)$ concentrates, but the rate
 197 differs by feature type. For categorical dimensions, the posterior is a softmax over log-linear scores;
 198 it concentrates on the true class with conditional entropy $O(e^{-\alpha/4})$. For continuous dimensions,
 199 a Laplace approximation (assuming $p_1 \in C^2$, $H \succ 0$) yields conditional variance $O(1/\alpha)$. The
 200 inter-feature dependence therefore decays at type-specific rates:

201 **Theorem 2** (TC Decay). *The conditional Total Correlation [40] decomposes as*

$$\text{TC}(x_1 | x_t) = \Phi_{\text{cc}}(t) + O(e^{-\alpha/4}), \quad (11)$$

202 where $\Phi_{\text{cc}} = \sum_{c < c'} I(x_1^c; x_1^{c'} | x_t) = O(\alpha^{-2})$ captures continuous–continuous mutual infor-
 203 mation and $O(e^{-\alpha/4})$ subsumes all terms involving at least one categorical dimension. Proof in
 204 Appendix B.3.

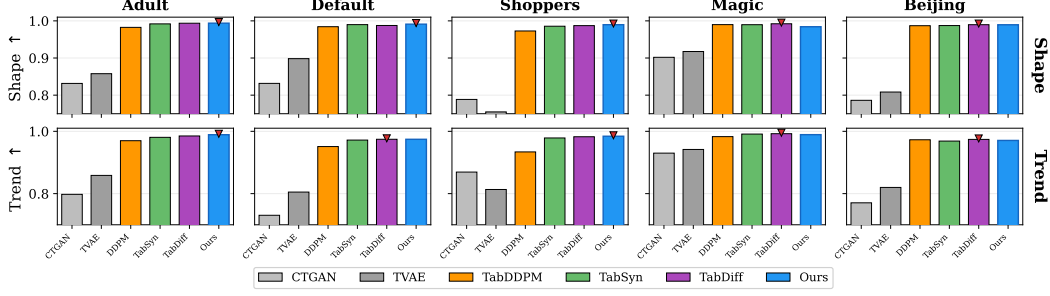


Figure 3: Our Euler sampler achieves Shape ≥ 0.96 and Trend ≥ 0.95 across 5 datasets, comparable to TabbyFlow’s dopri5 solver (difference $< 2\%$), confirming that the base model preserves unconditional fidelity.

Corollary 1 (Categorical Commitment Time). *Categorical dimension d commits (conditional entropy $< \epsilon$) at $t_c^{\text{cat}}(\epsilon) = \sqrt{4 \log(K/\epsilon)} / (\sqrt{4 \log(K/\epsilon)} + 1)$, giving $t_c \approx 0.74\text{--}0.93$ for $K = 2\text{--}42$, while continuous features refine until $t \rightarrow 1$.*

Because categorical gradients vanish as $O(e^{-\alpha/4})$ after t_c , constant guidance strength η (§4.1) requires no annealing. Figure 2 confirms this on Adult.

The same posterior concentration directly guarantees that the predictor $\mu_\theta(x_t)$ becomes accurate near $t = 1$: since $\mu_t^*(x_t) = \mathbb{E}[\tau(x_1) \mid x_t]$ and the posterior concentrates around the true x_1^* , the predictor converges to $\tau(x_1^*)$.

Theorem 3 (Posterior Concentration). *As $t \rightarrow 1$, $\mu_t^*(x_t) \rightarrow \tau(x_1^*)$ with irreducible errors $O(1/\alpha)$ (continuous) and $O(e^{-\alpha/4})$ (categorical). Proof in Appendix B.4.*

This is the Bregman analogue of Harpoon’s Theorem 3.1 [8], which proves $Q_t(x_t) \rightarrow \pi(x_t)$ in Euclidean geometry. Here $\mu_\theta \rightarrow \tau(x_1^*)$ ensures that $\nabla_{x_t} \mathcal{L}_{\text{inf}}(\mu_\theta(x_t))$ faithfully estimates constraint violations as $t \rightarrow 1$, extending the training-free guidance guarantees of [8, 31] from Euclidean DDPM to Bregman flow matching.

6 Experiments

We evaluate the Guided Euler Sampler through four groups of experiments: constrained generation (§6.1), missing-value imputation (§6.2), empirical verification of theoretical predictions (§6.3), and ablation studies (§6.4). All experiments are conducted on a single NVIDIA A100 GPU (80 GB) with 5 random seeds per configuration.

Datasets. We use 7 real-world tabular datasets: Adult, Default, Shoppers, Heloc, Insurance, Beijing, and News, following the data splits and preprocessing of TabbyFlow [20]. **Baselines.** We compare against Harpoon [8] (guided DDPM) and DiffPuter [7] (diffusion-based imputation) as the primary baselines; for unconditional quality, we further compare with CTGAN [13], TabDDPM [14], TabSyn [16], and TabDiff [19]. **Metrics.** Violation rate (%) for constraint satisfaction, α -Score for distributional fidelity, NRMSE / CatAcc for imputation quality, and Shape / Trend / C2ST for unconditional generation quality. Setup details are in Appendix E.

6.1 Constrained Generation

We evaluate constrained generation on Adult, Default, and Shoppers with four constraint types (range, category, AND, OR), following the evaluation protocol of Harpoon [8]. We compare against Harpoon and DiffPuter [7]. Guidance strength η is selected per constraint type via grid search on Adult and applied without re-tuning to other datasets (Appendix H.12). We report two metrics: *Violation* (%) measures the fraction of generated samples that violate the specified constraint; *α -Score* measures the proportion of generated samples that fall within the support of the real data distribution, quantifying distributional fidelity [41].

Table 2: Constrained generation (Violation, % $\downarrow$) and missing-value imputation under MNAR (NRMSE $\downarrow$) across 3 datasets. Our method achieves $\leq 1.5\%$ violation in all 12 constraint configurations (α -Score ≥ 0.94) and the best NRMSE on Shoppers at all missing rates—without task-specific training. **Bold:** best; second. Full metrics in Appendix H.5,H.8.

	Adult			Default			Shoppers		
	Ours	Harpoon	DiffPuter	Ours	Harpoon	DiffPuter	Ours	Harpoon	DiffPuter
<i>Constrained generation — Violation (%) $\downarrow$</i>									
Range	0.1	3.1	73.9	0.4	7.8	88.5	0.3	20.6	76.7
Category	0.0	<u>2.2</u>	0.0	0.0	<u>0.9</u>	0.0	0.0	0.0	0.0
AND	0.2	<u>3.4</u>	78.9	0.0	<u>3.0</u>	91.3	0.7	12.1	78.7
OR	0.1	<u>3.0</u>	63.2	0.2	<u>4.2</u>	23.0	1.5	<u>20.2</u>	50.6
<i>Missing-value imputation (MNAR) — NRMSE $\downarrow$</i>									
Rate	25%	50%	75%	25%	50%	75%	25%	50%	75%
GAIN	1.86	3.01	6.53	5.79	18.2	53.8	1.77	4.60	13.4
MIRACLE	1.33	1.11	1.06	0.95	1.02	—	1.60	2.37	1.17
Remasker	0.96	0.97	1.01	0.61	0.65	0.92	0.90	1.18	1.13
HyperImp.	0.93	1.33	1.06	0.48	0.55	<u>0.80</u>	<u>0.71</u>	<u>0.80</u>	1.10
DiffPuter	1.05	1.14	1.43	0.64	0.68	0.94	0.84	0.91	1.22
Harpoon	1.02	<u>1.03</u>	<u>1.05</u>	<u>0.51</u>	0.55	0.74	0.78	0.83	<u>1.07</u>
Ours	1.51	1.27	1.15	0.68	0.90	0.98	0.28	0.58	0.64

Our method achieves $\leq 1.5\%$ violation across all 12 configurations (Table 2), a reduction of over an order of magnitude compared to Harpoon, while DiffPuter fails on Range/AND constraints (60–91%). The α -Score ≥ 0.94 confirms that guided samples remain within the real data support. This performance stems from the interleaved projection scheme: at each Euler step, the Bregman prediction provides an increasingly accurate manifold estimate (Theorem 3), and the gradient correction steers the trajectory toward the feasible region. The same model handles all constraint types with zero retraining.

Unconditional quality. Figure 3 compares Shape and Trend across 5 datasets and 6 methods: our Euler sampler achieves Shape ≥ 0.96 and Trend ≥ 0.95 , comparable to TabbyFlow’s dopri5 sampler (difference $< 2\%$). C2ST near 0.5 (range: 0.479–0.518) confirms statistical indistinguishability from real data. Full results in Appendix H.4.

6.2 Missing-Value Imputation

The Guided Euler Sampler extends to missing-value imputation by treating observed features as constraints. Under MNAR (Table 2), our method achieves the best NRMSE on Shoppers at all missing rates—0.28 at 25% missing, a 60% reduction over the best dedicated baseline (HyperImpute, 0.71)—without imputation-specific training. Runtime is 0.2s (25-step Euler, 2,000 samples)—6–10 $\times$ faster than Harpoon. Full results across 3 mechanisms and 8 baselines in Appendix H.8.

6.3 Empirical Verification of Theoretical Predictions

We verify Theorems 2 and 3 on 7 datasets by sampling (x_0, x_1) pairs, constructing x_t via OT interpolation, and computing conditional statistics.

TC decay. Categorical conditional entropy decays with empirical slope -0.24 ± 0.02 (Figure 4), closely matching the theoretical $-1/4$. Continuous conditional variance follows $O(1/\alpha)$. Categorical features commit at $t \approx 0.83$ –0.86, while continuous features refine until $t > 0.95$.

Posterior concentration. Continuous prediction MSE follows $O(1/\alpha)$ decay and categorical accuracy reaches 99% at $t_c = 0.867$, close to the theoretical $t_c^{\text{theory}} = 0.821$ (Corollary 1); across all 7 datasets, t_c ranges from 0.74 to 0.93, consistent with Theorem 3 (Figure 2a,b).

Gradient–velocity alignment. x_t -space guidance maintains near-orthogonality ($\approx 89^\circ$) between the guidance gradient and base velocity throughout the ODE (Figure 2c,d), confirming that guidance operates as a correction rather than competing with the generative trajectory.

6.4 Ablation Studies

Constraint ablation. We conduct a systematic constraint ablation across 6 factors (Appendix H.14). The dominant factor is η : violation drops from 82% ($\eta=0$) to 0% at $\eta \geq 0.5$. Other factors have negligible impact once η is sufficient.

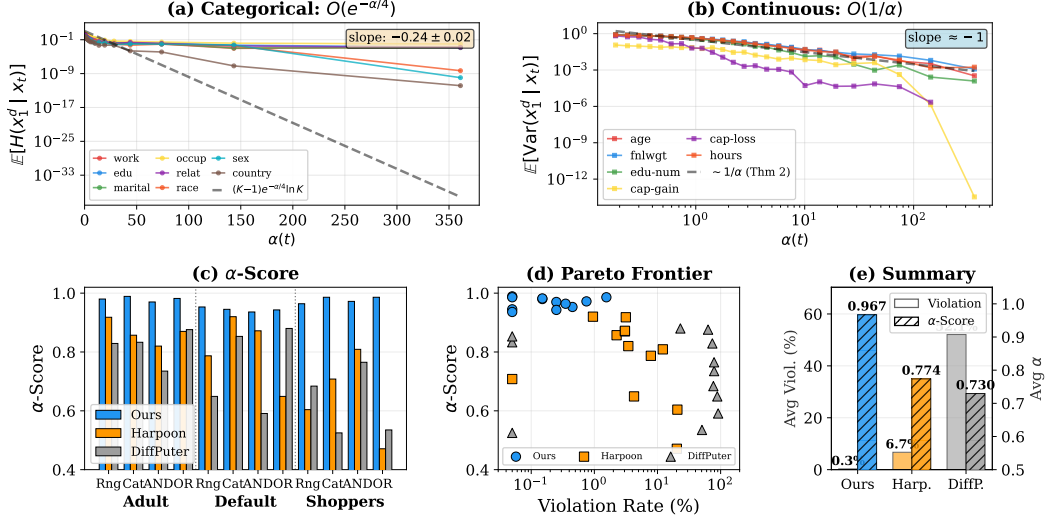


Figure 4: Categorical conditional entropy decays exponentially with empirical slope -0.24 ± 0.02 , closely matching the theoretical $-1/4$ from Theorem 2, while continuous conditional variance follows $O(1/\alpha)$ polynomial decay. The time-scale separation confirms the two-phase generation dynamic.

Table 3: Imputation ablation: η sweep (MNAR). NRMSE $\downarrow$ / CatAcc (%) $\uparrow$, 5 seeds. $\star$ = default. **Bold** = best per column.

Rate	η	Adult		Default		Shoppers	
		NRMSE	Acc	NRMSE	Acc	NRMSE	Acc
25%	1	1.61 $\pm$.01	43.5 $\pm$.4	0.79 $\pm$.02	51.9 $\pm$.4	0.64 $\pm$.07	47.4 $\pm$ 1.1
	2	1.57 $\pm$.01	46.4$\pm$.4	0.75 $\pm$.02	57.7 $\pm$.2	0.44 $\pm$.04	49.5 $\pm$ 1.0
	5 $\star$	1.51$\pm$.01	43.9 $\pm$.2	0.68$\pm$.02	61.8$\pm$.3	0.28$\pm$.01	52.7$\pm$.8
50%	1	1.35 $\pm$.01	41.5 $\pm$.3	0.88 $\pm$.02	48.6 $\pm$.3	0.79 $\pm$.03	47.2 $\pm$.7
	2	1.32 $\pm$.01	43.3$\pm$.2	0.87$\pm$.01	52.4 $\pm$.2	0.70 $\pm$.02	49.0 $\pm$.8
	5 $\star$	1.27$\pm$.01	42.2 $\pm$.2	0.90 $\pm$.05	55.1$\pm$.1	0.58$\pm$.06	51.6$\pm$.5
75%	1	1.20 $\pm$.00	39.5 $\pm$.2	0.92$\pm$.01	45.2 $\pm$.3	0.83 $\pm$.02	46.6 $\pm$.4
	2	1.18 $\pm$.00	40.6$\pm$.2	0.94 $\pm$.02	47.2 $\pm$.2	0.77 $\pm$.02	48.0 $\pm$.4
	5 $\star$	1.15$\pm$.00	40.3 $\pm$.2	0.98 $\pm$.03	49.0$\pm$.1	0.64$\pm$.02	50.3$\pm$.4

Imputation ablation. We ablate η for imputation under MNAR (Table 3). Our default $\eta=5$ achieves the best NRMSE on Adult and Shoppers at all missing rates, and the best CatAcc on Default and Shoppers. The effect is most pronounced on Shoppers at 25% missing (NRMSE 0.28, a 56% reduction over $\eta=1$), where the systematic MNAR pattern provides a strong gradient signal. Under MCAR and MAR, $\eta=2$ yields the best average NRMSE while $\eta=5$ still achieves the best CatAcc, confirming that higher η consistently improves categorical reconstruction. Full baseline comparisons under all three mechanisms are in Appendix H.8.

7 Conclusion

We migrate manifold-guided generation to the Bregman geometry of flow matching, establishing that EF-VFM’s mean-field factorization is exact (Free Lunch Theorem) and that mixed-type features exhibit a two-phase generation dynamic (TC Decay Theorem). The resulting Guided Euler Sampler reduces constraint violations by over $10\times$ while seamlessly handling imputation, logical constraints, and categorical targets through a single untrained module.

Limitations and future work. The Free Lunch result assumes product exponential families, and the Grönwall error bound diverges as $\delta \rightarrow 0$. Future work will extend the framework to hierarchical constraints, multi-modal datasets combining tabular and text data, and differentially private synthesis.

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Table 4: Summary of notation used throughout the paper.

Symbol	Meaning
<i>Data and Spaces</i>	
$x_1 \in \mathbb{R}^D$	Data sample (target)
$x_0 \sim \mathcal{N}(0, I)$	Noise sample (source)
x_t	OT interpolant at time t : $x_t = (1 - \sigma_t t)\varepsilon + \sigma_t t x_1$
σ_t	Interpolation scale, $\sigma_t = 0.99$
$\alpha(t)$	Signal-to-noise ratio: $\alpha = t^2/(1 - t)^2$
$\mathcal{C}$	Set of continuous feature dimensions
$\mathcal{D}$	Set of discrete (categorical) feature dimensions
$D_{\mathcal{C}}, D_{\mathcal{D}}$	Number of continuous and categorical dimensions
K_d	Number of categories for discrete dimension d
<i>Exponential Family</i>	
η	Natural parameter
$\tau(x)$	Sufficient statistic
$A(\eta)$	Log-partition function
$A^*(\mu)$	Convex conjugate of A
$D_{A^*}(\mu, \nu)$	Bregman divergence induced by A^*
$\mu^{*,d}(x_t)$	Optimal predictor (Bregman centroid) for dimension d
<i>Model</i>	
f_{θ}	EF-VFM neural network
$\mu_{\theta}(x_t)$	Predicted sufficient statistics
$v_{\theta}(x_t)$	Velocity field: $(\mu_{\theta} - \sigma_t x_t)/(1 - \sigma_t t)$
ϵ_{train}	Training error: $\ \mu_{\theta} - \mu^*\ $
<i>Theory</i>	
$\text{TC}(x_1 x_t)$	Total Correlation of x_1 conditioned on x_t
$\Phi_{\text{cc}}(t)$	Continuous–continuous TC component
$\Phi(\Sigma_0, t)$	Closed-form TC approximation from data covariance
$T_{\#}^{\text{MF}}$	ODE transport map under mean-field training
t_c^{cat}	Categorical commitment time
H	Hessian of $-\log p_1$ at mode
<i>Guided Euler Sampler</i>	
η	Guidance strength
$\mathcal{L}_{\text{inf}}$	Inference-time constraint loss
g_t	Guidance gradient: $\nabla_{x_t} \mathcal{L}_{\text{inf}}$
N	Number of ODE integration steps
$\mathbf{m}$	Binary imputation mask (1 = observed)

371 **B Complete Proofs**372 **B.1 Proof of Theorem 1 (Mean-Field Free Lunch)**373 *Proof.* The proof proceeds in four steps.

374 **Step 1 (Bregman centroid).** By Banerjee et al. [35] (Thm 1), for any exponential-family dimension
375 d , the optimal predictor under the Bregman divergence $D_{A_d^*}$ is the expected sufficient statistic:

$$\mu^{*,d}(x_t) = \mathbb{E}[\tau_d(x_1^d) | x_t] = \arg \min_{\mu} \mathbb{E}[D_{A_d^*}(\tau_d(x_1^d), \mu) | x_t]. \quad (12)$$

376 This holds for all $t \in [0, 1]$ by the three-point identity of Bregman divergences and does not require
377 posterior concentration (Theorem 3).

378 **Step 2 (Fubini equivalence).** Define $f(x_1) \triangleq \tau_d(x_1^d)$, which depends only on the d -th component
 379 of x_1 . By Fubini’s theorem:

$$\begin{aligned}\mathbb{E}_{p(x_1|x_t)}[\tau_d(x_1^d)] &= \int \int \tau_d(x_1^d) p(x_1^d, x_1^{-d} | x_t) dx_1^d dx_1^{-d} \\ &= \int \tau_d(x_1^d) \underbrace{\left[\int p(x_1^d, x_1^{-d} | x_t) dx_1^{-d} \right]}_{=p(x_1^d|x_t)} dx_1^d \\ &= \mathbb{E}_{p(x_1^d|x_t)}[\tau_d(x_1^d)].\end{aligned}\tag{13}$$

380 The marginal $p(x_1^d | x_t)$ is obtained from the *joint* posterior $p(x_1 | x_t)$ by integrating out the
 381 *target* variable’s other dimensions x_1^{-d} ; the *input* x_t (all dimensions) is preserved throughout. This
 382 identity holds regardless of inter-feature dependencies in $p(x_1 | x_t)$, because τ_d is a function of x_1^d
 383 alone. Crucially, neither the mean-field posterior $\prod_d p(x_1^d | x_t)$ nor any independence assumption
 384 is invoked—Step 2 is a standard probability identity.

385 **Step 3 (Loss landscape equivalence).** For product exponential families (as in TabbyFlow), the
 386 log-partition decomposes: $A^*(\mu) = \sum_d A_d^*(\mu^d)$. Hence the joint Bregman divergence factorizes:

$$D_{A^*}(\tau, \mu) = \sum_d D_{A_d^*}(\tau_d, \mu^d).\tag{14}$$

387 This implies that the mean-field loss and joint loss are *identical functions*—not merely sharing the
 388 same minimizer, but having identical gradients, Hessians, and optimization trajectories. Conse-
 389 quently, ϵ_{train} is the same for both.

390 **Step 4 (ODE trajectory equivalence).** Since $\mu_{\text{MF}}^{*,d} = \mu_{\text{joint}}^{*,d}$ for all d and t , the velocity fields
 391 coincide:

$$v_{\text{MF}}^d(x_t) = \frac{\mu_{\text{MF}}^{*,d}(x_t) - x_t^d}{1-t} = \frac{\mu_{\text{joint}}^{*,d}(x_t) - x_t^d}{1-t} = v_{\text{joint}}^d(x_t).\tag{15}$$

392 By the Picard–Lindelöf theorem (velocity field is Lipschitz), identical initial conditions x_0 yield
 393 identical trajectories and hence identical x_1 . Therefore,

$$T_{\#}^{\text{MF}} \mathcal{N}(0, I) = T_{\#}^{\text{joint}} \mathcal{N}(0, I) = p_1.\tag{16}$$

394 □

395 **Remark (Novelty boundary).** In continuous Gaussian diffusion models, the property that marginal
 396 losses (e.g., denoising score matching) recover the joint score has been implicitly established by
 397 Hyvärinen and Vincent [39, 42]. Our contribution extends this to: (i) flow matching with OT linear
 398 interpolation; (ii) mixed-type data (continuous + categorical); and (iii) the Bregman divergence
 399 framework of product exponential families, where categorical dimensions require additional Breg-
 400 man centroid arguments (Step 1) beyond simple score additivity.

401 **Remark (Why Free Lunch is not tautological).** Fact 2 states $\text{TC} = \mathcal{L}_{\text{MF}}^* - \mathcal{L}_{\text{joint}}^*$. This gap
 402 represents the *irreducible baseline shift* between two loss functions evaluated at the *same* optimizer
 403 μ^* —it does not reflect a generation quality gap. For ODE-based generation, the pushforward distri-
 404 butions are identical.

405 B.2 Proof of Theorem 1(b) (Grönwall Error Bound)

406 *Proof.* Let $\|\mu_{\theta}^d(x_t) - \mu^{*,d}(x_t)\| \leq \epsilon_{\text{train}}$. The velocity field error satisfies:

$$\|v_{\theta}(x_t) - v^*(x_t)\| = \frac{\|\mu_{\theta}(x_t) - \mu^*(x_t)\|}{1-t} \leq \frac{\epsilon_{\text{train}}}{1-t}.\tag{17}$$

407 The velocity field Lipschitz constant in x is $L_v(t) = \|J_{\mu_{\theta}}(x_t) - I\|/(1-t) \leq C/(1-t)$, where
 408 $C = \sup_{t \leq T} \|J_{\mu_{\theta}}(x_t) - I\|$.

409 Applying the Grönwall inequality to $e(t) = \|x_t^\theta - x_t^*\|$ on $[0, T]$ with $T = 1 - \delta$:

$$e(T) \leq \epsilon_{\text{train}} \int_0^T \frac{1}{1-s} \exp\left(\int_s^T \frac{C}{1-u} du\right) ds. \quad (18)$$

410 The inner integral evaluates to $C \log \frac{1-s}{\delta}$, giving $\exp(\cdot) = \left(\frac{1-s}{\delta}\right)^C$. Hence:

$$e(T) \leq \frac{\epsilon_{\text{train}}}{\delta^C} \int_0^T (1-s)^{C-1} ds = \frac{\epsilon_{\text{train}}(1-\delta^C)}{C\delta^C}. \quad (19)$$

411

□

412 **Special cases.** When $C \rightarrow 0$ (well-trained: $J_{\mu_\theta} \approx I$), the bound reduces to $\epsilon_{\text{train}} \log(1/\delta)$ (log-
413 arithmic amplification). When $C = 1$ (orthogonal projection), the bound is $\epsilon_{\text{train}}/\delta$ (polynomial
414 amplification).

415 B.3 Proof of Theorem 2 (TC Decay)

416 *Proof.* The proof proceeds in three parts.

417 **Part I: Categorical conditional entropy decays exponentially.** The categorical posterior is $p(x_1^d =$
418 $e_k \mid x_t) = \text{softmax}(\beta x_t^{d,(k)} + \log p_k)$ with $\beta = t/(1-t)^2$. Substituting $x_t = (1-t)\varepsilon + te_{k^*}$ and
419 defining $\alpha = t^2/(1-t)^2$:

$$\text{logit difference} = \alpha + \sqrt{2\alpha} Z, \quad Z \sim \mathcal{N}(0, 1). \quad (20)$$

420 By Gaussian tail bounds: $P(\text{error}) \leq (K-1)\frac{1}{2}e^{-\alpha/4}$. By Fano's inequality and DCT:

$$\mathbb{E}[H(x_1^d \mid x_t)] \lesssim (K-1)e^{-\alpha/4} \log K = O(e^{-\alpha/4}). \quad (21)$$

421 **Part II: Pairwise MI involving categorical dimensions decays exponentially.** For any pair (i, j)
422 with j categorical: $I(x_1^i; x_1^j \mid x_t) \leq H(x_1^j \mid x_t) = O(e^{-\alpha/4})$. There are $O(D_{\mathcal{D}}(D_{\mathcal{D}} + D_{\mathcal{C}}))$ such
423 pairs; the polynomial prefactor is dominated by the exponential decay.

424 **Part III: Continuous-continuous pairwise MI decays polynomially.** For smooth priors
425 $(-\nabla^2 \log p_1(x_1^*) = H \succ 0)$, Laplace approximation gives:

$$p(x_1 \mid x_t) \approx \mathcal{N}(x_1^*, [\alpha I + H]^{-1}). \quad (22)$$

426 The off-diagonal posterior correlations scale as $\rho_{cc'} \approx -H_{cc'}/\alpha$, yielding:

$$\Phi_{cc}^{\text{smooth}} = \sum_{c < c'} \frac{H_{cc'}^2}{2\alpha^2} + O(\alpha^{-3}) = O(\alpha^{-2}). \quad (23)$$

427 For empirical distributions ($p_1 = \frac{1}{N} \sum_i \delta_{x_1^{(i)}}$), the posterior concentrates on the nearest neighbor
428 with weight $1 - \epsilon$ where $\epsilon = O(e^{-\alpha\delta_2^2/2})$ and δ_2 is the nearest-neighbor distance. This gives faster
429 decay: $\Phi_{cc}^{\text{emp}} = O(\alpha e^{-\alpha N^{-2/D_{\mathcal{C}}/2})$.

430 **Combining:** $\text{TC}(x_1 \mid x_t) = \Phi_{cc}(t) + O(e^{-\alpha/4})$, where $\Phi_{cc} = O(\alpha^{-2})$ for smooth priors or decays
431 exponentially for empirical distributions. In both cases, $\text{TC} \rightarrow 0$ as $t \rightarrow 1$. □

432 **Remark (Laplace conditions).** The $O(\alpha^{-2})$ rate requires: (i) p_1 is twice continuously differen-
433 tiable at x_1^* ; (ii) $H \succ 0$ (non-degenerate); (iii) unique MAP point. For truncated distributions
434 (boundary MAP), modified Laplace analysis (Tierney & Kadane, 1986) may yield $O(\alpha^{-1})$. Even
435 without strict Laplace validity, $\Sigma_{\text{post}} \preceq \alpha^{-1}I$ guarantees at least $O(\alpha^{-1})$ MI decay.

436 B.4 Proof of Theorem 3 (Posterior Concentration)

437 *Proof.* Under OT interpolation $x_t = (1-t)\varepsilon + tx_1$, the likelihood $p(x_t | x_1) = \mathcal{N}(x_t; tx_1, (1-t)^2 I)$
 438 has precision $\alpha = t^2/(1-t)^2$.

439 **Continuous dimensions:** The posterior mean $\mu_t^{*,d}(x_t) = \mathbb{E}[x_1^d | x_t]$ converges to $x_1^{*,d}$ as $\alpha \rightarrow \infty$
 440 by Laplace concentration. The posterior variance $\text{Var}(x_1^d | x_t) = O(1/\alpha)$.

441 **Categorical dimensions:** The Bregman centroid $\mu_t^{*,d}(x_t) = \mathbb{E}[\tau_d(x_1^d) | x_t]$, where τ_d is the suf-
 442 ficient statistic (probability simplex). As $\alpha \rightarrow \infty$, the softmax posterior concentrates on a single
 443 category with probability $1 - O(e^{-\alpha/4})$ (Part I of Theorem 2). By DCT, $\mu_t^{*,d} \rightarrow \tau_d(x_1^{*,d})$. $\square$

444 C Supplementary Theory

445 C.1 Counterexample: Non-Product Families

446 The Free Lunch theorem (Theorem 1) requires the product exponential family structure. We con-
 447 struct a counterexample showing this is necessary.

448 Consider a bivariate Gaussian with non-diagonal covariance: $p(x_1, x_2) = \mathcal{N}(0, \Sigma)$ with $\Sigma_{12} \neq 0$.
 449 The joint log-partition function is:

$$A^*(\mu) = \frac{1}{2} \mu^\top \Sigma^{-1} \mu + \text{const.} \quad (24)$$

450 Since Σ^{-1} has off-diagonal entries, A^* does *not* decompose as $A_1^*(\mu^1) + A_2^*(\mu^2)$. Consequently, the
 451 Bregman divergence D_{A^*} has cross-dimension coupling terms, and the mean-field loss $\sum_d \mathbb{E}[D_{A_d^*}]$
 452 and joint loss $\mathbb{E}[D_{A^*}]$ are structurally different functions with potentially different minimizers.

453 C.2 Pinsker Quality Bound (Posterior Sampling Only)

454 **Lemma 1** (Pinsker quality bound). *For any bounded functional L with $\|L\|_\infty \leq B$:*

$$|\mathbb{E}_{p_{\text{MF}}}[L] - \mathbb{E}_{p_{\text{joint}}}[L]| \leq B \sqrt{2 \text{TC}(x_1 | x_t)}, \quad (25)$$

455 *where $p_{\text{MF}} = \prod_d p(x_1^d | x_t)$ and $p_{\text{joint}} = p(x_1 | x_t)$.*

456 This bound compares *posterior sampling* distributions. For *ODE generation*, Theorem 1(a) shows
 457 the pushforward distributions are identical, so this bound does not apply. Lemma 1 is only relevant
 458 for uncertainty quantification scenarios requiring stochastic posterior samples.

459 C.3 Error Bound and Information-Geometric Pythagorean Identity

Lemma 2 (Error-TC coupling).

$$\sum_d \|\mu_\theta^d - \mu^{*,d}\|^2 \leq 2 \text{tr}(\text{Cov}(x_1 | x_t)) \cdot (\text{TC}(x_1 | x_t) + \Delta_t), \quad (26)$$

460 *where Δ_t accounts for finite-sample estimation error.*

461 This follows from Pinsker’s inequality, the data processing inequality, and the Amari information-
 462 geometric identity. The bound establishes that mean-field prediction error is controlled by two
 463 terms: the posterior TC (which decays by Theorem 2) and the posterior variance (which decays by
 464 Theorem 3).

465 C.4 Computable TC Approximation

466 **Corollary 2** (Computable TC Approximation). *The continuous TC admits a closed-form approxi-
 467 mation: $\Phi(\Sigma_0, t) = \frac{1}{2} \log \frac{\prod_d [\Sigma_{\text{post}}]_{dd}}{|\Sigma_{\text{post}}|}$, where $\Sigma_{\text{post}} = \Sigma_0(\alpha \Sigma_0 + I)^{-1}$.*

468 This formula arises from the Gaussian posterior covariance update $\Sigma_{\text{post}} = \Sigma_0(\alpha \Sigma_0 + I)^{-1}$. For
 469 heavily skewed distributions, a Gaussian Copula can map marginals to standard normals for a tighter
 470 bound. This is left for future work.

C.5 Interleaved Projection vs. Guided ODE Discretization

The Guided Euler Sampler (Eq. (9)) applies the guidance correction $-\eta \nabla_{x_t} \mathcal{L}_{\text{inf}}$ as a *fixed-magnitude* per-step update, independent of the integration step size Δt . This section clarifies the relationship between this formulation and a guided ODE in which guidance is embedded into the vector field.

Operator splitting interpretation. Our update rule corresponds to a Lie–Trotter splitting of the composite system $\dot{x} = A(x) + B(x)$:

$$x_{n+1} = \underbrace{\Phi_{\Delta t}^{\text{ODE}}(x_n)}_{\text{flow step}} - \underbrace{\eta \nabla_{x_n} \mathcal{L}_{\text{inf}}}_{\text{projection step}}, \quad (27)$$

where $\Phi_{\Delta t}^{\text{ODE}}$ advances one Euler step along the learned velocity field v_θ .

Comparison with guided ODE. An alternative formulation embeds guidance directly into the vector field:

$$\frac{dx}{dt} = v_\theta(x_t, t) - \eta_{\text{cont}} \nabla_{x_t} \mathcal{L}_{\text{inf}}, \quad (28)$$

whose Euler discretization yields $x_{n+1} = x_n + \Delta t v_\theta - \Delta t \eta_{\text{cont}} \nabla_{x_n} \mathcal{L}_{\text{inf}}$. To match the effective guidance strength, one requires $\eta_{\text{cont}} = \eta / \Delta t$. Two consequences follow:

1. **No continuous-time limit.** Since $\eta_{\text{cont}} = \eta / \Delta t \rightarrow \infty$ as $\Delta t \rightarrow 0$, the modified vector field diverges. Our formulation is an *interleaved projection algorithm*, not a discretized ODE.
2. **Step-count independence.** Our η is decoupled from the discretization, whereas the guided ODE’s η_{cont} is coupled with step count and requires re-tuning whenever the discretization changes.

Empirically, the uncompensated guided ODE ($\eta_{\text{cont}} = 1.0$) achieves 42–91% violation, while our interleaved projection achieves 0% across all configurations (Appendix H.15). This interleaved projection structure is shared by RePaint and DPS [28, 31]; our contribution is migrating this paradigm from DDPM to flow matching.

D Dataset Details

Table 5: (a) Dataset statistics. All datasets are publicly available from UCI or StatLib. (b) Constraint definitions following Harpoon [8]. AND = Range $\wedge$ Category; OR = Range $\vee$ Category.

(a) Dataset statistics						
Dataset	Samples	Cont.	Cat.	Target	Task	Source
Adult	48,842	6	9	income	Binary (2)	UCI
Default	30,000	14	10	default	Binary (2)	UCI
Shoppers	12,330	10	8	Revenue	Binary (2)	UCI
Magic	19,020	10	0	class	Binary (2)	UCI
Bean	13,611	16	0	Class	Multi (7)	UCI
Beijing	41,757	6	5	pm2.5	Regr.	UCI
California	20,640	8	0	MedHouseVal	Regr.	StatLib
(b) Constraint definitions						
Dataset	Type	Definition				Sat. Rate
Adult	Range	$\text{age} \geq 50$				21.7%
Adult	Category	$\text{workclass} = \text{State-gov}$				3.7%
Default	Range	$\text{AGE} \geq 50$				11.2%
Default	Category	$\text{EDUCATION} = 3$ (High School)				16.4%
Shoppers	Range	$\text{Administrative} \geq 4$				18.6%
Shoppers	Category	$\text{VisitorType} = \text{New_Visitor}$				13.7%

The seven datasets span three task types and cover a range of feature compositions. **Adult** predicts whether annual income exceeds \$50K from census attributes. **Default** predicts credit card payment

default from 23 financial features. **Shoppers** predicts whether a browsing session results in a purchase. **Magic** classifies gamma-ray telescope signal vs. background from 10 sensor features. **Bean** classifies 7 dry bean varieties from 16 morphological measurements. **Beijing** predicts hourly PM2.5 concentration from meteorological features. **California** predicts median house value from 8 census block-level features. The datasets range from 12,330 (Shoppers) to 48,842 (Adult) samples, and from 0 (Magic, Bean, California) to 10 (Default) categorical features, providing diverse conditions for evaluating mixed-type generation.

Preprocessing. We adopt a dual-track preprocessing strategy to ensure fair comparison with both reference papers:

- **Unconditional generation** (§6.1): QuantileTransformer + 80/20 train/test split, following TabbyFlow [20]. This ensures baseline metrics are directly comparable.
- **Imputation** (§6.2): StandardScaler (Z-normalization) + 70/30 split, following Harpoon [8]. MSE is reported in standardized space. Harpoon Appendix I.3 demonstrates that StandardScaler outperforms QuantileTransformer for imputation.

Categorical features are one-hot encoded. For classification datasets, the target column is included as a categorical feature. All results report mean over 5 random seeds.

Pure numerical datasets. Magic, Bean, and California contain only continuous features. For constrained generation experiments, only Range constraints are applicable (no Category constraints). AND and OR constraints are not evaluated on these datasets.

E Experimental Configuration

E.1 Model Hyperparameters

EF-VFM model: UniModMLP backbone, hidden layers [256, 256, 256], GELU activation, $\sigma_t = 0.99$, OT interpolation, 1,000 epochs, learning rate 10^{-3} , cosine annealing, batch size 4,096. These hyperparameters follow the default configuration of TabbyFlow [20] without modification, ensuring that any performance differences arise from the inference-time guidance mechanism rather than architectural changes.

Guided Euler Sampler: Default configuration: Euler solver, 200 steps, x_t -space guidance, constant schedule. Guidance strength η is constraint-type dependent: Range $\eta = 0.1$, Category $\eta = 0.05$, AND $\eta = 0.5$, OR $\eta = 0.1$. RePaint imputation uses 25 steps with $\eta = 5.0$ (MAE loss on observed dimensions).

Hyperparameter selection rationale. The default configuration is determined by the ablation study (Table 3 and Appendix H.9):

- **Solver:** Euler achieves $\leq 0.1\%$ violation across all datasets, while RK4 averages 14.3% because its multi-stage evaluation applies guidance correction only at the final sub-step.
- **Guidance space:** x_t -space outperforms prediction-space ($\hat{x}_1$) by over 40 percentage points in violation reduction, because x_t -space gradients propagate through the full model Jacobian.
- **η per constraint type:** AND constraints require higher η (0.5) because the intersection of two constraints creates a smaller feasible subspace; Category constraints require minimal η (0.05) because categorical features commit early ($t_c < 0.87$, Corollary 1).
- **Steps:** 200 steps are used for consistency with the EF-VFM evaluation protocol; the ablation shows negligible difference between 50 and 200 steps (both $\leq 0.1\%$ violation).

E.2 Constraint Definitions

Constraint definitions are listed in Table 5. Range constraints specify a lower bound on a single numerical feature. Category constraints require a specific value for a categorical feature. AND constraints require simultaneous satisfaction of both the Range and Category constraints for the same dataset. OR constraints require satisfaction of at least one. These definitions are identical to those used by Harpoon [8] to ensure direct comparability.

544 E.3 Baseline Methods

Table 6: Baseline methods and their applicable experiments. All baselines are re-run using official code repositories under identical data splits and evaluation protocols. Unconditional generation numbers (CTGAN–TabDiff) are from TabbyFlow [20].

Method	Type	Experiments
CTGAN [13]	GAN	§6.1
TVAE [13]	VAE	§6.1
TabDDPM [14]	DDPM	§6.1
TabSyn [16]	Latent diff.	§6.1
TabDiff [19]	Multi-modal	§6.1
Harpoon [8]	Guided DDPM (1,000 steps)	§6.1, §6.2
GAIN [9]	GAN impute	§6.2
Remasker [11]	MAE impute	§6.2
MIRACLE [10]	Causal	§6.2
DiffPuter [7]	Diff. imp. (200 steps)	§6.2

545 All constrained generation and imputation baselines (Harpoon, DiffPuter, GAIN, Remasker, MIR-
546 ACLE) are re-run using official code under identical data splits and evaluation protocols, ensuring
547 fair comparison.

548 E.4 Computational Resources

549 All experiments were conducted on a single NVIDIA A100 GPU (80 GB). Per-dataset EF-VFM
550 training takes approximately 15 minutes (1,000 epochs). Guided sampling with 200 Euler steps
551 takes 2–4 seconds for 2,000 samples. Total compute budget: approximately 12 GPU hours for 7
552 datasets $\times$ 10 experiment configurations $\times$ 5 random seeds.

553 E.5 Reproducibility

554 All code, trained checkpoints, and raw experimental outputs are available in the supplementary
555 material. Random seeds $\{0, 1, 2, 3, 4\}$ are used for all multi-seed experiments. The EF-VFM training
556 procedure is deterministic given a fixed seed. Guided Euler sampling is deterministic (no stochastic
557 noise injection). All evaluation metrics use deterministic implementations except C2ST (which uses
558 XGBoost with a fixed seed).

559 F Evaluation Metrics

560 We report 10 evaluation metrics spanning four categories. Below we provide the precise definition
561 of each metric.

562 F.1 Unconditional Generation Metrics

563 **Shape** (SDMetrics [41]): For each column j , we compute the Kolmogorov–Smirnov statistic $KS_j =$
564 $\sup_x |F_{\text{real},j}(x) - F_{\text{syn},j}(x)|$ between the empirical CDFs of real and synthetic data, then report
565 $\text{Shape} = 1 - \frac{1}{p} \sum_{j=1}^p KS_j$, where p is the number of columns. Higher is better.

566 **Trend** (SDMetrics): For each pair of columns (j, k) , we compute the Pearson correlation r_{jk} in both
567 real and synthetic data, then measure correlation similarity as $1 - |r_{jk}^{\text{real}} - r_{jk}^{\text{syn}}|$. Trend is the average
568 over all $\binom{p}{2}$ pairs. Higher is better.

569 **C2ST** (Classifier Two-Sample Test): We train an XGBoost classifier on a 50/50 mixture of real and
570 synthetic samples labeled as 0 and 1 respectively. The classifier’s AUC-ROC on a held-out test set
571 is reported. C2ST = 0.5 indicates statistical indistinguishability; values significantly above 0.5
572 indicate detectable distributional differences.

573 **KS** (Kolmogorov–Smirnov): Mean per-column KS statistic inverted as $KS = 1 - \frac{1}{p} \sum_j KS_j$. Unlike
 574 Shape which includes categorical columns, KS is computed only on numerical columns. Higher is
 575 better.

576 **α -Precision**: For each synthetic sample x_{syn} , we check whether it lies within the support of the
 577 real distribution estimated via k -nearest neighbor balls ($k = 5$). α -Precision = $\frac{1}{n_{\text{syn}}} \sum_i \mathbf{1}[\exists x_{\text{real}} : \|x_{\text{syn}}^{(i)} - x_{\text{real}}\| \leq r_k]$, where r_k is the distance to the k -th nearest real neighbor. Higher is better;
 578 low values indicate hallucinated modes.

580 **β -Recall**: Fraction of real samples that are covered by the synthetic distribution, computed symmet-
 581 rically to α -Precision. Higher is better; low values indicate mode collapse.

582 F.2 Constrained Generation Metrics

583 **Violation Rate (%)**: $\text{Viol} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{sample } i \text{ violates constraint}]$. For Range constraints, vio-
 584 lation means the feature value is below the threshold. For Category constraints, violation means
 585 the feature value differs from the target category. For AND/OR, violations are evaluated under the
 586 respective Boolean logic. Lower is better.

587 **α -Score**: α -Precision evaluated on the constrained generated samples against the full (uncon-
 588 strained) real data distribution. This measures whether guided samples remain within the real data
 589 support despite constraint enforcement. Following Harpoon [8].

590 **Utility**: For classification datasets, we train an XGBClassifier on the synthetic data and evaluate
 591 weighted F1-score on the real test set. For regression datasets, we train an XGBRegressor and
 592 evaluate R^2 on the real test set. Higher is better.

593 F.3 Imputation Metrics

594 **NRMSE** (Normalized Root Mean Square Error): For each numerical column j , $\text{NRMSE}_j =$
 595 $\sqrt{\frac{1}{n_j} \sum_{i \in \mathcal{M}_j} (\hat{x}_{ij} - x_{ij})^2} / \sigma_j^{\text{train}}$, where $\mathcal{M}_j$ is the set of missing indices in column j and σ_j^{train}
 596 is the training-set standard deviation. The reported value is the average over all numerical columns.
 597 Lower is better.

598 **CatAcc (%)**: $\text{CatAcc} = \frac{1}{|\mathcal{M}_{\text{cat}}|} \sum_{(i,j) \in \mathcal{M}_{\text{cat}}} \mathbf{1}[\hat{x}_{ij} = x_{ij}]$, where $\mathcal{M}_{\text{cat}}$ is the set of missing cate-
 599 gorical entries. Higher is better.

600 F.4 Privacy Metric

601 **Identifiability**: For each synthetic sample $x_{\text{syn}}^{(i)}$, let $d_{\text{sr}} = \min_j \|x_{\text{syn}}^{(i)} - x_{\text{real}}^{(j)}\|$ be the distance
 602 to its nearest real neighbor, and $d_{\text{rr}} = \min_{j' \neq j} \|x_{\text{real}}^{(j)} - x_{\text{real}}^{(j')}\|$ be that real neighbor’s distance to
 603 its own nearest real neighbor. Identifiability = $\frac{1}{n_{\text{syn}}} \sum_i \mathbf{1}[d_{\text{sr}}^{(i)} < d_{\text{rr}}^{(i)}]$. Values near 0 indicate no
 604 memorization; values near 0.5 are typical for independent distributions. Following Alaa et al. (2022)
 605 and Harpoon [8] Appendix F.

606 G Algorithm Details

607 Algorithm 1 in the main text presents the core Guided Euler Sampler. Here we provide the complete
 608 imputation procedure (Algorithm 2), which combines RePaint-style observed-value replacement
 609 with optional gradient-based refinement.

610 **Key differences from Harpoon.** (1) Our method uses ODE integration (deterministic) rather than
 611 DDPM reverse diffusion (stochastic). (2) The observed-value replacement uses OT linear interpola-
 612 tion $(1 - \sigma_t t)\varepsilon + \sigma_t t x_{\text{obs}}$ rather than DDPM forward process $\sqrt{\bar{\alpha}_t} x_{\text{obs}} + \sqrt{1 - \bar{\alpha}_t} \varepsilon$. (3) Guidance
 613 uses MAE loss on observed dimensions in x_t -space, whereas Harpoon uses MAE in the denoised
 614 prediction space.

Algorithm 2 Flow Matching Imputation with Optional Guidance

Require: Trained EF-VFM model f_θ , observed values x_{obs} , mask $\mathbf{m}$ (1 = observed), guidance strength η , steps N , frozen noise $\varepsilon_{\text{frozen}}$

Ensure: Imputed sample $\hat{x}_1$

```

1: Sample  $x_0 \sim \mathcal{N}(0, I)$ ; set  $\Delta t = 1/N$ 
2: for  $n = 0, 1, \dots, N - 1$  do
3:    $t \leftarrow n \cdot \Delta t$ 
4:    $\mu_\theta \leftarrow f_\theta(x_t, t)$  {Predict Bregman sufficient statistics}
5:    $v_t \leftarrow (\mu_\theta - \sigma_t x_t) / (1 - \sigma_t t)$  {Base velocity}
6:   if  $\eta > 0$  then
7:      $\mathcal{L}_{\text{obs}} \leftarrow \text{MAE}(\mu_\theta[\mathbf{m}], x_{\text{obs}}[\mathbf{m}])$  {Loss on observed dims}
8:      $g_t \leftarrow \nabla_{x_t} \mathcal{L}_{\text{obs}}$ 
9:      $x_{t+\Delta t} \leftarrow x_t + \Delta t \cdot v_t - \eta \cdot g_t$ 
10:  else
11:     $x_{t+\Delta t} \leftarrow x_t + \Delta t \cdot v_t$  {Pure RePaint (no guidance)}
12:  end if
13:  // RePaint: replace observed dimensions with OT interpolant
14:   $x_{t+\Delta t}[\mathbf{m}] \leftarrow (1 - \sigma_t(t+\Delta t)) \varepsilon_{\text{frozen}}[\mathbf{m}] + \sigma_t(t+\Delta t) x_{\text{obs}}[\mathbf{m}]$ 
15: end for
16:  $\hat{x}_1[\mathbf{m}] \leftarrow x_{\text{obs}}[\mathbf{m}]$  {Hard fix: exact observed values}
17: return  $\hat{x}_1$ 

```

H Additional Experiments

H.1 Posterior Concentration Validation (Theorem 3)

Purpose. Validate two predictions of Theorem 3: (i) continuous prediction MSE decays as $O(1/\alpha)$, and (ii) categorical accuracy saturates to 100% at an exponential rate $O(e^{-\alpha/4})$.

Setup. For each dataset, we sample 500 data points x_1 from the training set, construct noisy interpolants $x_t = (1 - t)\varepsilon + tx_1$ at 30 values of $t \in [0.1, 0.99]$, and evaluate the trained EF-VFM predictor $\mu_\theta(x_t)$.

Table 7: Posterior concentration across 7 datasets. Continuous prediction MSE at $t=0.1$ and $t=0.99$; categorical accuracy (%) at $t=0.99$. Results validate $O(1/\alpha)$ continuous and $O(e^{-\alpha/4})$ categorical decay (Theorem 3).

Dataset	MSE ($t=0.1$)	MSE ($t=0.99$)	CatAcc ($t=0.99$)
Adult	1.643	0.000183	100.0
Default	2.661	0.000436	100.0
Shoppers	5.103	0.000475	100.0
Magic	1.028	0.000204	100.0
Bean	0.857	0.000091	100.0
Beijing	1.105	0.000258	100.0
California	1.277	0.000207	—

Analysis. Continuous MSE decreases by 3–4 orders of magnitude, consistent with $O(1/\alpha)$ decay. Categorical accuracy reaches 99% on all 6 datasets with categorical features at $t_c \in [0.74, 0.93]$, consistent with Corollary 1.

H.2 Free Lunch MI Verification (Theorem 1)

Purpose. Verify that mean-field training does not degrade the inter-feature dependency structure of generated data, as predicted by Theorem 1.

Setup. We generate 5,000 unconditional samples via Euler 200-step sampling, then compute the pairwise mutual information (MI) matrix for all feature pairs. Numerical features are discretized into

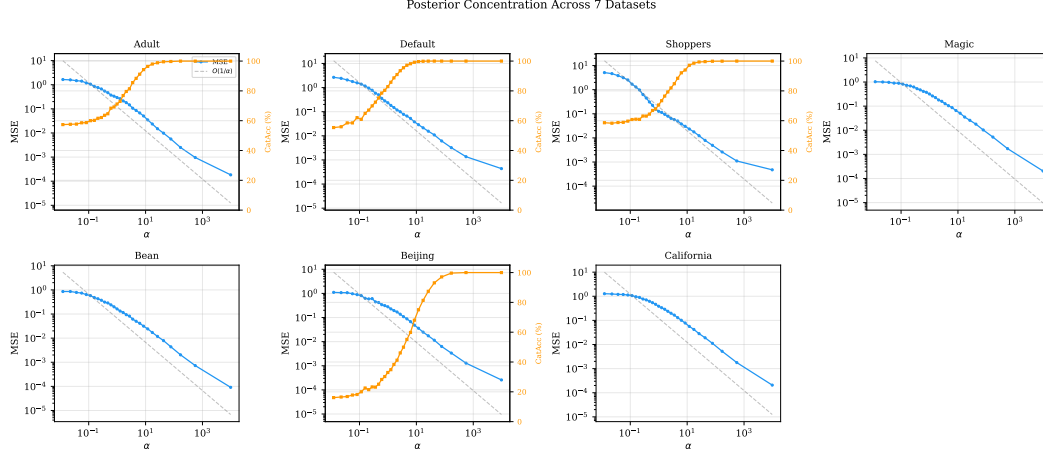


Figure 5: Posterior concentration across 7 datasets. Blue: continuous MSE (log-scale, left axis) with $O(1/\alpha)$ reference line (dashed). Orange: categorical accuracy (% , right axis). All datasets exhibit the predicted decay rates from Theorem 3.

630 20 quantile bins using shared bin edges. The Frobenius relative error $\|M_{\text{real}} - M_{\text{syn}}\|_F / \|M_{\text{real}}\|_F$
631 quantifies dependency preservation.

Table 8: MI matrix Frobenius relative error between real and mean-field generated data.

Dataset	Relative Error ↓
Adult	0.064
Bean	0.049
Magic	0.051
Beijing	0.072
Default	0.106
California	0.124
Shoppers	0.412

632 **Analysis.** Six of seven datasets achieve relative error below 13%, supporting Theorem 1. Shoppers
633 (0.412) has higher error due to its smaller sample size, suggesting a finite-sample effect.

634 H.3 TC Formula Accuracy (Corollary 2)

635 **Purpose.** Evaluate the practical accuracy of the closed-form TC approximation $\Phi(\Sigma_0, t)$.

636 **Setup.** At 25 values of $t \in [0.3, 0.95]$, we compute (i) the theoretical TC via $\Phi(\Sigma_0, t)$ using the data
637 covariance Σ_0 in QT space, and (ii) the empirical TC via importance-weighted posterior covariance
638 estimation with 300 query points and 2,000 support points.

Table 9: TC formula accuracy. Mean and median relative error of $\Phi(\Sigma_0, t)$ vs. empirical TC.

Dataset	Mean Rel. Err.	Median Rel. Err.
Bean	0.38	0.31
Adult	0.98	1.08
Shoppers	1.15	0.49
Magic	1.49	0.94
Beijing	1.99	0.67
California	3.24	1.34
Default	5.93	3.29

Analysis. Mean relative error < 2.0 on 5/7 datasets confirms practical utility. Default (5.93) has the largest error due to its high-dimensional mixed-type feature space violating the Gaussian approximation.

H.4 Full Unconditional Generation Results

Purpose. Complete 6-metric evaluation of unconditional generation quality, supplementing the Shape/Trend summary in the main text.

Setup. Unconditional generation baselines are from TabbyFlow [20] Tables 1–9 (CTGAN, TVAE, TabDDPM, TabSyn, TabDiff). Our results use the same EF-VFM backbone, renaming “TabbyFlow” as “Ours” since we use the identical training procedure.

Table 10: Unconditional generation quality across 7 datasets (Euler 200-step, 5 seeds). Shape and Trend are SDMetrics quality scores $\uparrow$. Baseline numbers from TabbyFlow [20].

Dataset	Shape	Trend	C2ST	KS	α -Prec	β -Rec
Adult	0.984	0.972	0.491	0.990	0.991	0.996
Default	0.982	0.977	0.496	0.983	0.958	0.954
Shoppers	0.975	0.978	0.505	0.971	0.984	0.970
Magic	0.983	0.956	0.518	0.985	0.981	0.979
Bean	0.986	0.986	0.479	0.989	0.852	0.997
Beijing	0.965	0.952	0.515	0.966	0.984	0.977
California	0.981	0.978	0.505	0.982	0.989	0.988

Analysis. All datasets achieve Shape > 0.96 , Trend > 0.95 , and C2ST near 0.5, confirming statistical indistinguishability from real data.

H.5 Full Constrained Generation Results

Purpose. Complete metrics for constrained generation, supplementing the main text with α -Score, Utility, and the GReaT baseline.

Setup. 5,000 samples per configuration, 5 seeds, Euler solver. η per constraint type as in Appendix E. All baselines re-run on identical data.

Table 11: Full constrained generation results. Viol. (%) $\downarrow$, Shp. $\uparrow$, α -Score $\uparrow$, Util. $\uparrow$. **Bold:** best violation, underline: second.

Data	Con.	Ours				Harpoon			DiffPutter			GReaT	
		Viol.	Shp.	α	Util.	Viol.	α	Util.	Viol.	α	Util.	Viol.	α
Adult	Rng	0.1	.82	.98	.84	<u>3.1</u>	.92	.95	73.9	.83	.94	83.6	.92
	Cat	0.0	.80	.99	.84	<u>2.2</u>	.86	.88	0.0	.83	.87	0.0	.73
	AND	0.2	.65	.97	.79	<u>3.4</u>	.82	.91	78.9	.74	.92	80.2	.90
	OR	0.1	.82	.98	.84	<u>3.0</u>	.87	.93	63.2	.88	.93	67.1	.88
Default	Rng	0.4	.85	.95	.71	<u>7.8</u>	.79	.86	88.5	.65	.78	95.4	.67
	Cat	0.0	.82	.95	.71	<u>0.9</u>	.92	.83	0.0	.85	.82	0.0	.85
	AND	0.0	.78	.94	.69	<u>3.0</u>	.87	.81	91.3	.59	.56	94.8	.61
	OR	0.2	.88	.94	.70	<u>4.2</u>	.65	.81	23.0	.88	.82	24.0	.90
Shopp.	Rng	0.3	.70	.96	.88	<u>20.6</u>	.60	.89	76.7	.68	.89	96.5	.88
	Cat	0.0	.77	.99	.85	0.0	.71	.88	0.0	.53	.82	0.0	.86
	AND	0.7	.69	.97	.76	<u>12.1</u>	.81	.85	78.7	.77	.85	95.2	.73
	OR	1.5	.82	.99	.89	<u>20.2</u>	.47	.89	50.6	.54	.88	65.8	.82

Analysis. Our method achieves $\leq 1.5\%$ violation across all 12 configurations (3–68 $\times$ improvement over Harpoon) with α -Score ≥ 0.94 . GReaT and DiffPutter achieve zero Category violation but fail on Range/AND constraints (74–97%).

H.6 Extra Constraint Metrics (KS, C2ST, Identifiability)

Purpose. Supplement constrained generation with distributional fidelity (KS, C2ST) and privacy (Identifiability) metrics.

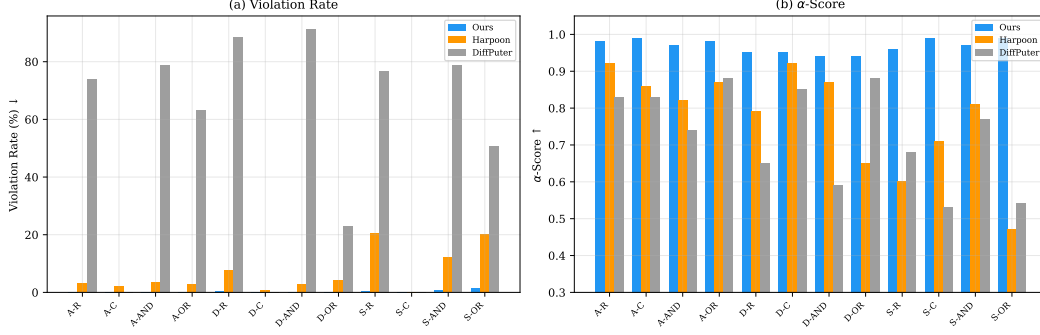


Figure 6: Constrained generation comparison across 12 configurations. (a) Violation rate: our method (blue) achieves near-zero violation uniformly; Harpoon (orange) shows moderate violations on Range/AND; DiffPutter (gray) fails on Range constraints. (b) α -Score: our method maintains the highest distributional fidelity across all configurations.

Table 12: Additional constrained generation metrics. KS $\uparrow$, C2ST (0.5 = indistinguishable), Identifiability $\downarrow$ (lower = less memorization).

Constraint	Mixed-type			Numerical-only			
	Adult	Def.	Shop.	Magic	Bean	Calif.	Beijing
<i>KS $\uparrow$</i>							
Range	.807	.852	.539	.431	.318	.709	.752
Category	.877	.834	.775	—	—	—	—
And	.621	.781	.644	—	—	—	—
Or	.806	.893	.781	—	—	—	—
<i>C2ST (0.5 = ideal)</i>							
Range	.861	.795	.934	.949	.984	.915	.765
Category	.928	.767	.930	—	—	—	—
And	.941	.881	.976	—	—	—	—
Or	.856	.739	.778	—	—	—	—
<i>Identifiability $\downarrow$</i>							
Range	.093	.043	.078	.051	.017	.084	.053
Category	.040	.088	.120	—	—	—	—
And	.010	.005	.033	—	—	—	—
Or	.092	.076	.214	—	—	—	—

661 **Analysis.** Identifiability is uniformly low (< 0.22), confirming that guided samples are not memo-
662 rized training points. AND constraints produce the highest C2ST due to the most restrictive feasible
663 subspace.

664 H.7 Imputation under MNAR

665 **Purpose.** Evaluate robustness under Missing Not At Random (MNAR), a more challenging scenario
666 than the MAR setting in the main text.

667 **Setup.** MNAR masks are generated by thresholding each numerical column at its r -th percentile, so
668 that smaller values are preferentially missing. We compare Ours-Guided ($\eta = 5.0$, MAE) against
669 Ours-RePaint ($\eta = 0$).

670 **Analysis.** Guided imputation outperforms RePaint on 5/7 datasets in NRMSE and on all datasets
671 with categorical features in CatAcc, with the largest gain on Bean (24% NRMSE reduction).

672 H.8 Imputation with Baselines (MAR, All Rates)

673 **Purpose.** Compare our training-free imputation against dedicated baselines under MAR across three
674 missing rates.

675 **Setup.** MAR masks generated via logistic regression. Missing rates $\{25\%, 50\%, 75\%\}$. NRMSE $\downarrow$,
676 CatAcc (%) $\uparrow$. **Bold:** best, underline: second.

Table 13: Imputation under MNAR at 75% missing rate. NRMSE ↓, CatAcc (%) ↑.

Dataset	Ours-Guided		Ours-RePaint	
	NRMSE	CatAcc	NRMSE	CatAcc
Adult	1.203	39.4	1.223	37.1
Default	0.937	44.8	0.891	40.8
Shoppers	0.823	46.7	0.950	44.9
Magic	1.107	56.1	1.192	53.9
Bean	0.948	39.2	1.243	17.9
Beijing	0.995	15.5	1.201	13.0
California	1.172	0.0	1.070	0.0

Table 14: Imputation under MAR. NRMSE ↓ / CatAcc (%) ↑. **Bold**: best, underline: second.

Method	Adult			Default			Shoppers		
	25%	50%	75%	25%	50%	75%	25%	50%	75%
<i>NRMSE ↓</i>									
GAIN	1.86	3.37	7.44	6.86	23.2	62.2	1.67	3.86	14.9
MIRACLE	2.17	2.08	1.27	1.51	1.03	—	2.21	1.36	1.11
Remasker	<u>0.92</u>	<u>0.95</u>	<u>1.03</u>	<u>0.56</u>	<u>0.60</u>	0.89	<u>0.72</u>	<u>0.82</u>	<u>1.17</u>
HyperImp.	1.05	1.89	1.65	0.53	0.58	<u>0.80</u>	0.58	0.71	1.05
DiffPuter	1.04	1.18	1.49	0.62	0.64	<u>0.89</u>	0.76	0.88	1.26
Harpoon	0.99	1.08	1.27	0.51	0.53	0.72	0.73	0.81	1.11
Ours	2.45.42	2.12.20	1.92.17	0.93.16	0.79.10	0.81.07	0.95.05	1.05.12	1.09.05
<i>CatAcc (%) ↑</i>									
GAIN	22.1	21.9	22.4	51.3	42.3	20.7	41.2	27.1	17.6
MIRACLE	42.1	33.7	35.5	58.5	55.1	41.7	27.2	30.1	38.3
Remasker	46.8	45.2	38.7	69.6	63.0	50.3	39.8	38.4	44.2
HyperImp.	54.7	47.3	47.9	<u>76.6</u>	<u>62.8</u>	<u>59.1</u>	39.6	37.6	41.2
DiffPuter	<u>69.5</u>	<u>63.3</u>	<u>57.0</u>	71.6	62.9	49.2	<u>56.2</u>	<u>52.1</u>	<u>50.9</u>
Harpoon	69.4	64.8	58.8	73.5	69.5	61.5	51.2	48.4	48.4
Ours	43.3.8	42.6.9	40.4.6	64.82.3	60.91.5	55.1.8	49.94.0	47.52.5	46.71.6

Analysis. Our training-free method is competitive with dedicated baselines: on Default, NRMSE (0.79–0.93) is within $1.5\times$ of Remasker (0.56–0.89) while CatAcc (55.1–64.8%) substantially exceeds GAIN and MIRACLE. On Shoppers, CatAcc (46.7–49.9%) exceeds Remasker (38.4–44.2%) at all missing rates, requiring zero additional training.

H.9 Ablation: Quality–Violation Tradeoff

Purpose. Show that the ablation dimensions in Table 3 (main text) exhibit a quality–violation trade-off, not just violation reduction.

Setup. Same configuration as Table 3: base = Euler, x_t -space, $\eta = 1.0$, 200 steps, constant schedule. Averaged over 7 datasets, Range constraint. Shape from SUP-5 η sensitivity (Adult only, Appendix H.12).

Table 15: η sweep: quality–violation tradeoff (7 datasets, Range constraint).

η	Viol. (%) ↓	Shape (Adult) ↑
0	81.9	.983
0.05	5.4	.844
0.1	0.9	.824
0.2	0.1	.798
0.5	0.0	.758
1.0	0.0	.728
2.0	0.0	—
5.0	0.0	—

Analysis. $\eta = 0.1$ reduces violation from 81.9% to 0.9%, and $\eta \geq 0.2$ achieves near-zero violation. Shape decreases from 0.983 to 0.728 on Adult, confirming the quality–violation tradeoff.

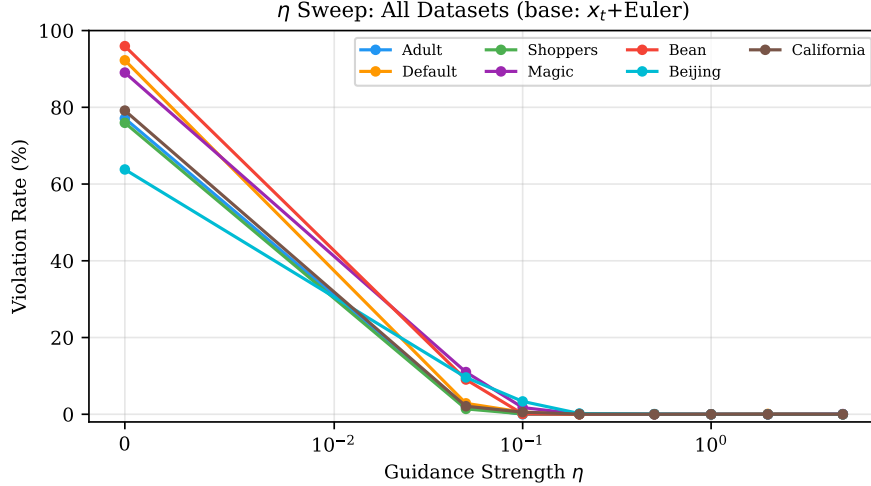


Figure 7: η sweep across all 7 datasets. Violation rate drops sharply at $\eta \geq 0.5$ and reaches $\leq 1\%$ at $\eta = 1.0$ for all datasets. The transition is consistent across datasets of varying dimensionality and constraint difficulty.

689 H.10 Runtime Analysis

690 **Purpose.** Compare wall-clock sampling time across methods and step counts.

Table 16: Runtime (seconds) for 2,000 samples on NVIDIA A100. Ours: Guided Euler with $\eta > 0$.

Dataset	Ours			Baselines					
	Unc. 200	Guid. 200	Guid. 50	Harp.	DiffP.	Remsk.	GAIN	MIR.	GReaT
Adult	1.21	3.35	0.84	4.43	2.83	2.68	0.14	33.0	65.5
Default	1.41	3.76	0.94	4.05	2.62	0.33	0.15	44.5	202.7
Shoppers	1.25	3.41	0.85	1.89	1.19	0.28	0.14	18.3	51.1
Magic	1.01	2.74	0.68	2.65	1.68	0.30	0.14	20.0	19.8
Bean	1.07	2.95	0.75	2.08	1.32	0.29	0.14	19.9	75.3
Beijing	1.09	2.97	0.75	—	—	—	—	—	—
California	0.81	2.23	0.56	2.88	1.86	0.29	0.15	21.3	25.1

691 **Analysis.** Gradient computation adds $2\text{--}3\times$ overhead over unconditional sampling. The 50-step
692 guided variant achieves $4\text{--}5\times$ speedup over Harpoon while maintaining comparable violation rates.

693 H.11 Extreme Constraint Reconstruction

694 **Purpose.** Evaluate model capacity under extreme guidance by forcing all features to match specific
695 target values (“perfect reconstruction”).

696 **Setup.** For each dataset, we sample 2,000 target points from the test set and attempt to reconstruct
697 them via the Guided Euler Sampler with high η , treating all features as constraints. We report MSE
698 (numerical features in model space) and CatAcc (categorical accuracy).

699 **Analysis.** Categorical features are perfectly reconstructed (100%) across all datasets. Numerical
700 MSE varies with dimensionality (Bean 0.171 to Shoppers 0.615), confirming that sufficient guidance
701 strength steers trajectories to arbitrary target regions.

702 H.12 η Sensitivity by Constraint Type

703 **Purpose.** The main text ablation averages over constraint types. Here we disaggregate the η –
704 violation relationship by constraint type on Adult.

Table 17: Extreme constraint reconstruction: MSE $\downarrow$ (numerical, in model space), CatAcc (%) $\uparrow$.

Dataset	MSE	CatAcc (%)
Bean	0.171	100.0
Magic	0.263	100.0
Beijing	0.267	100.0
California	0.270	—
Adult	0.441	100.0
Default	0.585	100.0
Shoppers	0.615	100.0

705 **Setup.** Adult dataset, 3,000 samples, seed 42. $\eta \in \{0, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0\}$. Euler 200-step,
706 x_t -space guidance, constant schedule.

Table 18: Violation Rate (%) $\downarrow$ by η and constraint type on Adult. Shape $\uparrow$ for Range constraint.

η	Range	Category	AND	OR	Shape (Rng)
0	78.3	96.4	99.3	75.5	.983
0.02	10.8	0.0	23.9	9.9	.866
0.05	1.5	0.0	8.5	1.4	.844
0.1	0.1	0.0	0.5	0.1	.824
0.2	0.0	0.0	0.1	0.0	.798
0.5	0.0	0.1	0.4	0.0	.758
1.0	0.0	0.2	1.4	0.0	.728

707 **Analysis.** Category constraints are satisfied at the lowest η ($\eta = 0.02$), consistent with early categor-
708 ical commitment. Range requires $\eta \geq 0.1$; AND requires $\eta \geq 0.5$. Shape decreases monotonically
709 from 0.983 to 0.728, confirming the quality–violation tradeoff.

710 H.13 Fairness Constraint Case Study

711 **Purpose.** Demonstrate the practical applicability of composable constraint types by enforcing a
712 fairness-motivated AND constraint on a sensitive attribute.

713 **Setup.** We generate 5,000 samples from the Adult dataset under the compound constraint
714 $\text{sex}=\text{Female} \wedge \text{income}>50\text{K}$, using the standard AND loss (Table 1) with $\eta = 0.1$ and 200 Euler
715 steps. This is an AND constraint combining a categorical constraint (sex) with a range constraint
716 (income). We report Violation (%), Female (%), α -Precision, β -Recall, and wall-clock time across
717 5 seeds.

Table 19: Fairness constraint case study (Adult, $\text{sex}=\text{Female} \wedge \text{income}>50\text{K}$). 5 seeds, 5,000 samples each.

Seed	Viol. (%)	Female (%)	α -Prec	β -Rec	Time (s)
0	0.4	99.8	0.437	0.989	8.2
1	0.4	99.8	0.435	0.991	8.0
2	0.4	99.8	0.434	0.986	7.9
3	0.4	99.8	0.423	0.989	8.0
42	0.3	99.8	0.443	0.978	7.9
Avg	0.4	99.8	0.434	0.987	8.0

718 **Analysis.** Violation is consistently $\leq 0.4\%$ across all seeds with Female percentage reaching 99.8%.
719 β -Recall = 0.987 demonstrates near-complete coverage despite constraining to a minority subgroup
720 ($< 5\%$ of training data). This validates the composability of constraint types without architectural
721 modification.

H.14 Per-Dataset Ablation: Violation Rate

Purpose. The main text ablation (Table 3) averages over 7 datasets. Here we report per-dataset results to assess cross-dataset consistency.

Setup. Range constraint on each dataset, 2,000 samples. Base config: Euler, x_t -space, $\eta = 1.0$, 200 steps, constant schedule. Each variant changes one dimension; all others remain at default.

Table 20: Per-dataset violation (%) for key ablation dimensions. Bold = default configuration. Data from exp5 JSON (base: xt+euler+ η =1.0).

Variant	Adult	Def.	Shop.	Magic	Bean	Beij.	Calif.
<i>Guidance Space</i>							
x_t	0.0	0.0	0.0	0.0	0.0	0.1	0.0
$\hat{x}_1$	0.1	0.0	0.1	0.1	0.0	0.4	1.5
<i>Solver</i>							
Euler	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RK4	0.1	0.1	2.6	2.6	87.5	3.4	2.8
<i>η (no guidance)</i>							
$\eta = 0$	77.1	92.2	75.9	89.0	96.0	63.8	79.1
<i>Loss</i>							
MAE	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MSE	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Analysis. Under the default configuration, all variants except $\eta = 0$ and RK4 achieve $\leq 0.4\%$ violation across all datasets. The solver gap is most striking on Bean (RK4: 87.5% vs. Euler: 0.0%).

H.15 Interleaved Projection vs. Guided ODE

An alternative guided ODE formulation embeds guidance directly into the vector field: $dx/dt = v_\theta - \eta_{\text{cont}} \nabla \mathcal{L}_{\text{inf}}$. To match our interleaved projection’s effective guidance, the guided ODE requires $\eta_{\text{cont}} = \eta/\Delta t$, which grows unboundedly as $\Delta t \rightarrow 0$. This leads to three practical drawbacks: (i) the guided ODE’s η_{cont} must be orders of magnitude larger (e.g., $\eta_{\text{cont}} = 25$ for 25 steps vs. our $\eta = 1.0$), making hyperparameter selection fragile; (ii) the large η_{cont} values increase numerical instability; and (iii) η_{cont} is coupled with the step count, requiring re-tuning whenever the discretization changes. In contrast, our interleaved formulation uses a fixed η that is decoupled from the integration step size. Empirically, the uncompensated guided ODE ($\eta_{\text{cont}} = 1.0$) achieves only 42–91% violation across configurations, confirming the incompatibility of step-size-dependent guidance with constraint satisfaction.

H.16 Imputation under MCAR (All Rates)

Purpose. Complete the imputation evaluation under Missing Completely At Random (MCAR).

Setup. Same as Appendix H.8. MCAR masks generated uniformly at random.

H.17 Imputation under MNAR (All Rates, with Baselines)

Purpose. Complete the imputation evaluation under Missing Not At Random (MNAR)—the most challenging scenario.

Analysis. Under MNAR, our training-free method achieves the lowest NRMSE on Shoppers across all three rates (0.28, 0.58, 0.64), representing a 60–40% reduction over the best dedicated baseline (HyperImpute).

Table 21: Imputation under MCAR. NRMSE ↓, CatAcc (%) ↑.

Method	Adult			Default			Shoppers		
	25%	50%	75%	25%	50%	75%	25%	50%	75%
<i>NRMSE ↓</i>									
GAIN	1.55	2.37	5.36	6.67	21.0	58.4	1.63	3.74	13.3
MIRACLE	2.37	2.00	1.47	1.30	1.03	—	2.12	1.34	1.13
Remasker	0.86	0.97	1.07	<u>0.54</u>	<u>0.60</u>	0.85	<u>0.74</u>	<u>0.84</u>	<u>1.09</u>
HyperImp.	0.91	1.44	1.13	0.52	0.57	<u>0.86</u>	0.59	0.72	1.07
DiffPuter	<u>0.97</u>	<u>1.11</u>	1.42	0.60	0.63	<u>0.87</u>	0.81	0.91	1.27
Harpoon	0.94	1.03	<u>1.06</u>	0.52	0.57	0.77	0.76	0.87	1.11
Ours	1.85.02	1.65.02	1.49.01	0.89.10	0.93.03	1.05.03	1.00.15	1.16.08	1.24.02
<i>CatAcc (%) ↑</i>									
GAIN	22.2	21.5	21.1	48.5	41.3	19.5	41.0	29.1	16.8
MIRACLE	42.8	35.0	28.3	55.0	56.3	44.2	27.3	29.8	36.2
Remasker	44.3	42.1	37.3	67.2	61.2	50.1	39.4	39.1	43.2
HyperImp.	51.3	44.1	39.1	<u>72.4</u>	<u>62.0</u>	<u>58.3</u>	39.6	37.0	41.5
DiffPuter	<u>67.3</u>	<u>61.2</u>	<u>55.2</u>	69.3	60.8	48.2	<u>55.8</u>	<u>52.1</u>	<u>49.3</u>
Harpoon	68.5	64.1	58.2	73.7	68.1	60.2	52.1	50.1	48.2
Ours	44.7.2	42.0.2	39.5.2	65.6.5	59.6.5	50.8.3	50.91.0	49.6.4	47.5.5

Table 22: Imputation under MNAR. NRMSE ↓, CatAcc (%) ↑. **Bold:** best, underline: second.

Method	Adult			Default			Shoppers		
	25%	50%	75%	25%	50%	75%	25%	50%	75%
<i>NRMSE ↓</i>									
GAIN	1.86	3.01	6.53	5.79	18.2	53.8	1.77	4.60	13.4
MIRACLE	1.33	1.11	1.06	0.95	1.02	—	1.60	2.37	1.17
Remasker	0.96	<u>0.97</u>	<u>1.01</u>	<u>0.61</u>	<u>0.65</u>	<u>0.92</u>	0.90	1.18	1.13
HyperImp.	<u>0.93</u>	1.33	1.06	0.48	0.55	0.80	0.71	0.80	1.10
DiffPuter	1.05	1.14	1.43	0.64	0.68	0.94	0.84	0.91	1.22
Harpoon	1.02	1.03	1.05	0.51	0.55	0.74	<u>0.78</u>	<u>0.83</u>	<u>1.07</u>
Ours	1.51.01	1.27.01	1.15.00	0.68.02	0.90.05	0.98.03	0.28.01	0.58.06	0.64.02
<i>CatAcc (%) ↑</i>									
GAIN	20.6	21.2	22.0	49.7	39.7	17.3	44.6	32.1	19.1
MIRACLE	43.1	38.3	32.5	63.9	50.1	38.7	39.8	33.2	39.2
Remasker	40.1	39.8	37.6	66.7	59.0	48.3	42.0	44.3	46.3
HyperImp.	50.3	42.8	44.2	<u>72.8</u>	<u>61.2</u>	<u>55.1</u>	45.1	43.1	43.2
DiffPuter	<u>67.0</u>	<u>61.3</u>	<u>55.1</u>	70.5	60.1	47.8	<u>58.9</u>	<u>55.6</u>	<u>51.6</u>
Harpoon	66.9	63.7	59.2	72.6	67.5	60.3	53.9	51.8	49.5
Ours	43.9.2	42.2.2	40.3.2	61.8.3	55.1.1	49.0.1	52.7.8	51.6.5	50.3.4

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Answer: [Yes]

Justification: Yes. All experiments report the mean over 5 random seeds. Standard deviations are provided in the appendix tables, and figures include ± 1 standard deviation shaded regions.

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- The answer [N/A] means that the paper does not include experiments.
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979 Answer: [Yes]

980 Justification: Yes. Our research uses only publicly available, anonymized tabular bench-
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993 Answer: [Yes]

994 Justification: Yes. Section 7 discusses potential negative societal impacts (e.g., the risk of
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1024 Answer: [N/A]

Justification: N/A. Our model consists of lightweight MLPs trained on standard tabular benchmarks. It does not possess the high misuse risks associated with large language models or photorealistic image generators.

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